Will Bridging The Digital Divide Bring Prosperity To The Second "Next Billion Users" And Their Home Countries?

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GitHub: https://github.com/Merrimack-Capstone/MSDS-capstone

Abstract

History has shown that disadvantaged populations experience improved wellbeing when infrastructure improvements impact their lives. A diverse set of stakeholders, including governments, aid organizations and private sector technology companies, would benefit from being able to quantify (at minimum) and predict (ideally) the extent to which expanded access to the internet is one such infrastructure improvement that leads directly to higher population wellbeing. This report describes in depth the application of statistical and machine-learning based predictive analytics to both correlate and isolate causal links between access to the internet and population wellbeing. Using 30 years worth of data, consisting of a subset of raw, transformed and abstracted data vectors from the World Bank (for inputs that could influence wellbeing), and the United Nations (for the Human Development Index chosen as the basis for measuring wellbeing), this report demonstrates that the correlation between internet access and population wellbeing is positive and statistically relevant. This report also demonstrates that, while it is challenging to firmly quantify a causal relationship between internet access and wellbeing, well-performing broader models that predict wellbeing rely on internet access as an important positive factor in predicted wellbeing. Finally,this report shows that investing in broadening access to (ideally, renewable) electricity in developing countries is arguably the MOST important factor influencing wellbeing, as well as a necessary precursor to broadening internet access as a wellbeing growth accelerant. These findings can serve as support for prescriptive strategy for both private and public sector entities. Furthermore, because the data acquisition modules and predictive models that are the backbone of this study are built on public data and can be updated dynamically as new data becomes available, entities relying on these models to support their strategy can update the results regularly and make interpretative course-corrections as indicated.

Background & Question

The Rural Electrification Act ("REA"), passed in the United States as part of President Roosevelt's "New Deal" legislation in 1936, was designed to bring electricity (initially) and other infrastructure (later) to large portions of the country that were considered by industry to be unprofitable places for infrastructure investment. By all accounts, it was a rousing success because "the act allowed the U.S. to grow rapidly in the postwar period and to attain the position of economic dominance it now enjoys…by investing in the development of the country…(it) provided an economic springboard that propelled America ahead of other nations from the 1930s onward." (Taylor, 2024)

Globally, over the last three decades, the expansion of internet connectivity and digital technologies has presented a parallel opportunity to address long-standing global disparities in livelihood, economic participation, and education. While a significant portion of the world's population has been online for several decades, efforts such as Google's Next Billion Users initiative have been focused on developing applications and ways of connecting with the internet for those who reside in under-developed countries. These users are in the developmental stages of global digital inclusion and often access the internet through devices with less computational power, reliability and connection speed, if they can access it at all (Ranjan, 2022).

This study examines the relationship between digital inclusion and the wellbeing of emerging user populations. We employ both supervised and unsupervised machine learning and predictive modeling methods to attempt to correlate growth in digital inclusion with improved population wellbeing in "clusters" of countries / populations. We further attempt to establish a causal relationship between growth in digital inclusion and population wellbeing. In the absence of proof of such a clear predictive relationship, we also explore how important a role digital inclusion plays in predictive models that can successfully predict population wellbeing using a broader set of predictors.

Stakeholders

There are four distinct sets of stakeholders who will benefit from this project:

- Disadvantaged populations
- Public sector leaders of underdeveloped nations
- Private sector leaders in technology and e-commerce
- Global aid organizations

Research Question

This report addresses the following research question: *"To what extent has growth in internet access of populations influenced population wellbeing in more developed nations, and can that be used to predict improvements in wellbeing and economic progress among global minority populations (the "next billion users")?"*

This topic is not novel in and of itself. A number of studies have attempted to quantify this relationship - for example, one study produced a correlation between a country's % of population accessing the internet and that country's wellbeing - defined using the United Nation's Human Development Index (HDI - see "Data Acquisition" for further discussion) - with a 0.77 R^2 (Pratama and Al-Shaikh, 2012), and Jyad et al, while another study suggests that HDI can be predicted with a high accuracy (R^2 of 0.94) using internet users, broadband access and electricity as predictors (Jyad et al, 2021). However, we seek to build on and improve these studies in two key ways:

- Previous studies do not explore *causation* between internet access and wellbeing, whereas our study explores causation through the use of lagged predictors and panel regression
- Previous studies are dated and reflect much smaller sample sizes (in time period and countries) and do not include the effect of the explosion in internet access tied to the Covid 19 pandemic - our study is far more expansive

Data Acquisition

Data Acquisition : This study uses the following data sets, with predominantly automated data acquisition as noted (see Appendix A for a full data dictionary) - all data is imported directly into R dataframes:

- **PREDICTORS:** World Bank: Global Development Indicators
  - *Contains*: A subset of development indicators by country and year, going back to 1995, for well over 200 countries
  - *Acquisition Method:* Spreadsheet download from World Bank (Income Groupings); API calls directly retrieve data from the World Bank (all other data)
- **RESPONSE:** UNDB: Wellbeing and income inequality indicators
  - *Contains:* The HDI index of wellbeing, the Gini index of income inequality, and the IHDI inequality- adjusted wellbeing index, reported annually by country.
  - *Acquisition Method:* API calls directly retrieve data from UNDB

Since the API calls take 15-20 minutes to complete, imported data is saved in R ".rds" files, which are then imported into the Exploratory Data Analysis (EDA) code module. In order to update the data, the API data calls must be re-run to create updated static files.

Data Cleaning

The following steps were taken to prepare the raw data for EDA :

- Reduced broad set of 1,500 potential predictors within 88 separate "topics" (categories) to 40 variables we considered potentially relevant to population wellbeing through manual examination
- Dropped data prior to 1995 (pre-internet) as irrelevant to our study

- Identified "InternetUsersPct" as the most relevant predictor measuring digital inclusion, which is at the heart of our study.
- Identified HDI Index[1] and Adjusted HDI Index[2] as potential responses variables, but then discarded Adjusted HDI due to data sparseness
- Standardized the country codes in all three data sets
- Joined the World Bank, UNDB data and Income Group data using country code

Data Exploration

We generated a range of tables and visualizations to explore the data and inform feature selection, engineering, dimensionality reduction, and model design. These outputs, provided in significant detail in Appendix B, cover the following categories:

- ***Descriptive Statistics:*** Basic tabular analysis of every variable was used to understand data distribution, range and missing values. Tables 1 below shows an excerpt from these statistics which are explored fully in Appendix B
- ***Data Visualizations:*** We used histograms, violin plots, scatterplots, bar charts, and heatmaps to visualize relationships and correlations between variables, data distribution / skewness challenges, potential clustering / control variables, and data sparseness / missingness, and to perform an initial basic test of whether our hypothesis has a chance of being proven.
- ***Correlation Analysis:*** We used visualized (heatmap) and numeric correlation analysis tools, both filtered and unfiltered, to explore the correlations between features, both to identify features that are likely to be predictive and to identify potential collinearity, which can guide our

[1] HDI and its derivatives are development indices created and tracked by the United Nations which measure the level of development of a country using three primary inputs - population income levels, health (measured by longevity expectations at birth), and education.
[2] HDI index, adjusted for income inequality

decisions on features to drop and/or combine to reduce dimensionality and reduce the impact of collinearity on statistical model variance.

- ***Missing Data Exploration:*** we calculated the proportion of missing data across two axes (by variable, and by year) and visualized the results. These visualizations help identify patterns in missing data to guide decisions on how to fill in (e.g. interpolate missing years). They also helped guide decisions on which variables and/or entire year sets may simply be too sparse and should be dropped entirely/
- **Illustrations:** Figures 1 through 5 are examples of these tables and visualizations. The full set of exhibits, along with the insights gained which were used to guide our pre-processing and feature engineering, can be found in Exhibit B.

Figure 1

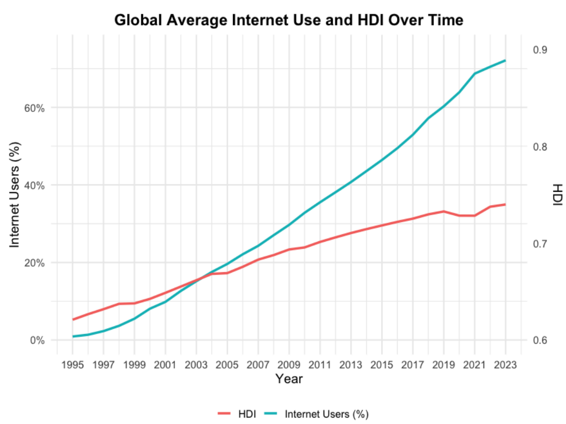


Figure 2

Figure 3

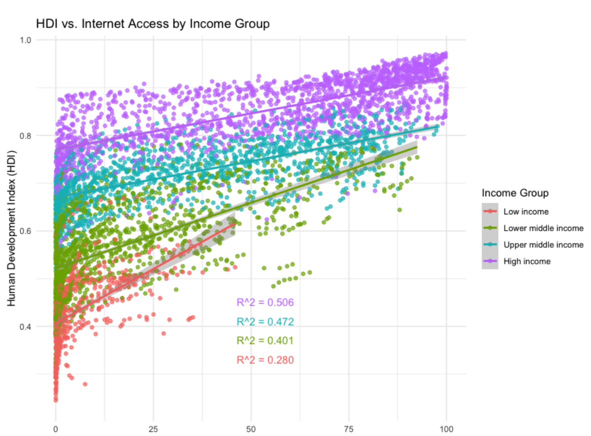


Figure 4

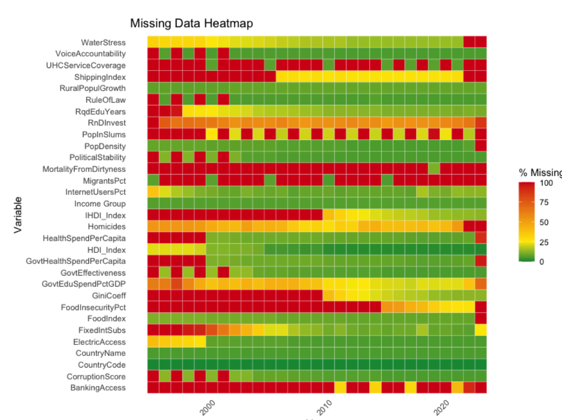
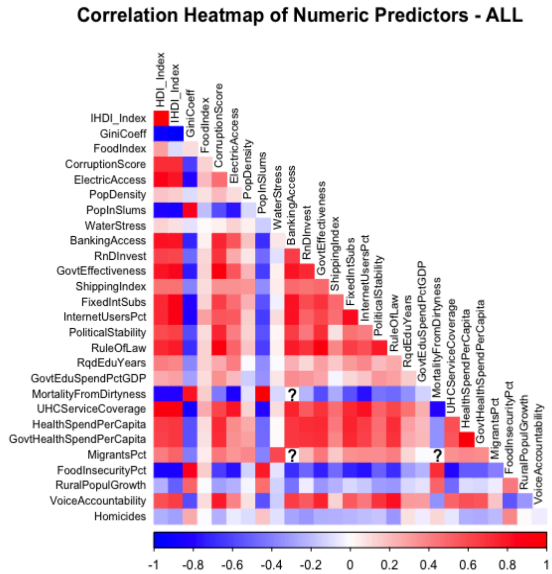


Figure 5



Summary of Findings from Exploration:

- Internet Usage and HDI (Figures 1 and 2) are positively correlated
- The consistently low R^2 values across income clusters (Figure 3) highlight a key limitation: internet access alone is not a strong enough predictor of overall wellbeing.
- There is a substantial amount of missing data that will result in dropped observations and/or data imputation. (Figure 4 above). Figure 4 and Figures 10-12 of Appendix B show that some variables and years exhibit such extensive missingness that they should be dropped entirely.
- It's also evident that there's a general trend of improved data completeness over time, and that there are certain patterns in missingness that will guide our strategy for dealing with null values (e.g. by interpolating values when data is only available every other year).
- There is significant collinearity between potential predictors (Figure 5 above and Figures 7-9 of Appendix B). This will need to be addressed in pre-processing through feature dropping, combining, and other dimensionality / collinearity reduction techniques

- There is a pronounced left skew in the InternetUsersPct data, with a large cluster of data points at or near 0% internet access (Figures 3 and 4 of Appendix B). This will need to be addressed in preprocessing via transformation.
- There are pronounced differences in data distribution by Income Group (Figures 5 and 6 Appendix B), suggesting that segmenting data by income group may improve models.
- Time series analyses (Figures 13-15 of Appendix B) reveal insights which at a minimum reinforce that internet access alone cannot be used to predict wellbeing, and may even suggest that our thesis is flawed, at least for the very low income countries which motivate this study. A deeper analysis of these charts may suggest that for very low income countries, internet access is simply not an important driver of HDI, and that for countries with significant income inequality, the benefits of improved internet access, along with other economic benefits, accrue to a too-narrow subset of the population.

Models

Following (and informed by) the EDA, we built and evaluated a series of linear regression models as well as machine learning models (both tree-based and neural network versions) to explore the relationship between and predictive value of internet usage and HDI. The modeling process was done in two stages:

Stage 1: Pre-processing and dimensionality reduction / feature engineering[3]

As a necessary precursor to building sound models, we implemented the following processes to prepare the data:

[3] This report contains a summarized explanation of the pre-processing we performed to develop our final feature and response set. A full review of the step by step process we followed, with extensive visualizations, can be seen by running our "Preprocessing" R code module

Feature Creation

We created two new features that represent the year-on-year change in internet access and HDI, in order to use changes rather than absolute levels in our models. We also created 1 and 2 year lagged Internet Usage variables as well as a 3 year cumulative change Internet Usage feature, to incorporate the effect of time lag and explore causality.

Initial Data Dropping

We dropped data, guided by heat maps of data "missingness" (both overall across variables and years) for variables, years and countries with more than 40% of their data missing Figures 6 and 7 show some of the initial cross-sectional heatmaps used to drive this initial broad-based data dropping.

Figure 6

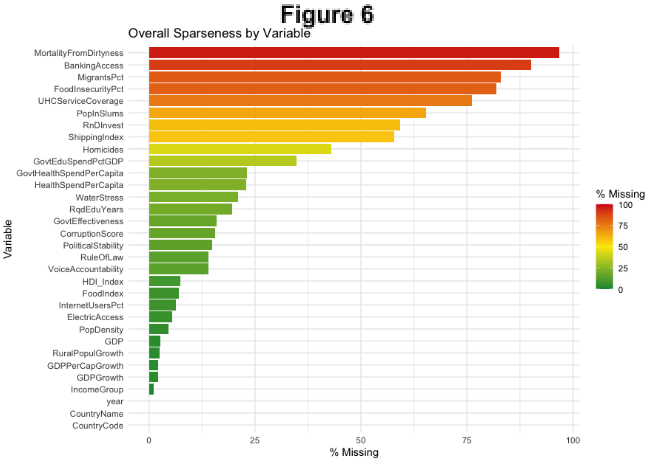
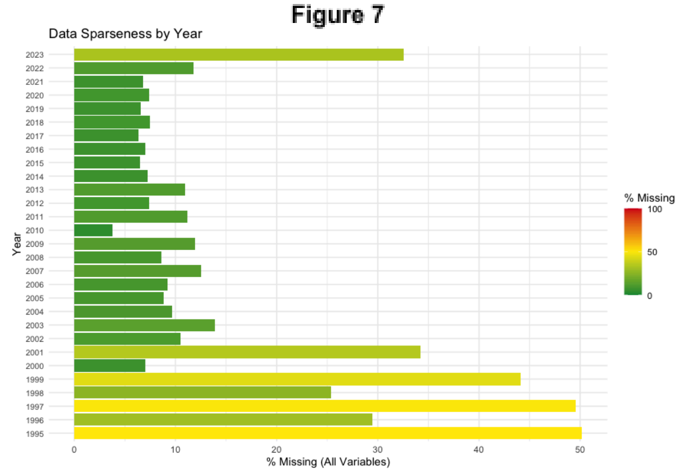


Figure 7



Training / Test Data Split Considerations

We considered how to split the data into training and test sets using a 70/30 ratio:

- Time series panel data such as that in our study presents particular issues when splitting because much of the data is "autocorrelated", meaning that a particular observation is likely to be highly correlated to observations immediately preceding or following it.
- Since our time series panel data is by country and year, to avoid autocorrelation risk, we elected to keep entire countries' data together, and split the countries up 70/30 with 70% of the countries appearing in the training data and the remaining 30% in the training data.

- Data split was done AFTER imputation of missing values (for reasons outlined below), but BEFORE Yeo Johnson transforms and Principal Component Analysis (since those two algorithms must be applied to the training data, with resulting parameters applied to the test data)
- To avoid bias, we stratified the countries by IncomeGroup so that the same rough distribution by IncomeGroup was maintained in both data sets - this was done based on the EDA analysis that showed distinct differences in HDI and internet access based on Income Group.

Data Imputation for missing values in the entire data set

Figure 8

Since all imputation was done intra-country (see below) using other time periods for the same country, and each country's entire data set is self-contained within either training or test data, there is no risk of leakage by doing intra-country data imputation on the entire data set pre-split. The revised missing data heatmap (Figure 8) was used as a blueprint to determine the process we used for imputation, which followed this process:

- Linear interpolation:
  - Voice Accountability, RuleOfLaw, PopInSlums, Political Stability, GovtEffectiveness and CorruptionScore
  - Later years of UHCServiceCoverage:

- Where data was only reported every other year (later years)
- Larger data gaps in earlier years between 2000 and 2015, where data was reported only every five years, but only after demonstrating that the the trajectory of the data over those twenty years was a virtually straight line, justifying that approach.

■ Missing 2001 "governance" variables which were oddly not reported - again justifiable given the short period spanned.

- Quadratic extrapolation: Missing data for 2023 using all the prior country/year pairs
- All other data was imputed based on the availability of other intra-country data for that variable as follows:[4]:
    - *3 or more available data points:* Quadratic regression-based imputation
    - *If 2 available data points:* Linear regression-based imputation
    - *1 available data point was available:* Data point replication
    - *0 available data points:* Left NA

Skewness

Yeo-Johnson transformation was applied to address skewness in the internet usage features (see results on the training data in Figure 9)

[4] It's important to note that while this approach is less defensible, the missingness it was applied to was quite sporadic, and without it, dropping rows with NA values before modeling would have cut our data set more than in half. This is true except for GovtEduSpendPctGDP which had more substantial missing data (25%), but that we felt merited salvaging given that education is one of three components of our HDI response variable and this is one of only two education-related predictors we have.

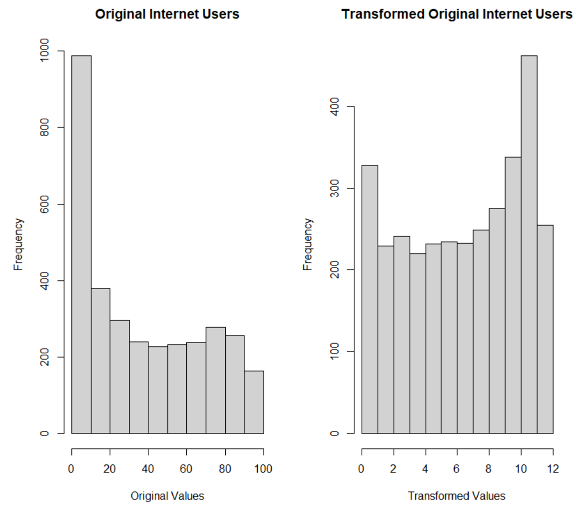


Figure 9
Results of Yeo Johnson Transform
Training Data

Binning

New binned fields were created for several reasons as follows:

- Collinear variables related to government accountability were combined using their mean
- Collinear variables related to public health initiatives were combined using their mean
- A binned HDI Category variable was created using UNDB's rules that rank countries' human development as "Low", "Medium", "High," and "Very High"

Unsupervised learning

Principal Component Analysis (PCA) was applied to the training data in order to help reduce dimensionality, reduce noise, and speed up the training of learning models by simplifying the dataset without losing significant information. As can be seen in Figure 10, 70% of the variance across the predictors can be explained using the top 5 Principal Components ("PC's), which reduces model dimensionality and collinearity.

Those top 5 PC's were stored as additional predictors in the training data set for potential modeling use. In addition, the "loadings" created by the PCA on the training data were applied to the test data to create comparable PC features.

Figure 10

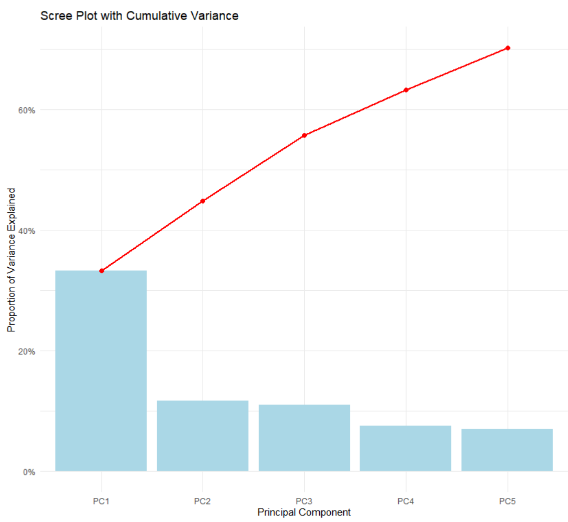
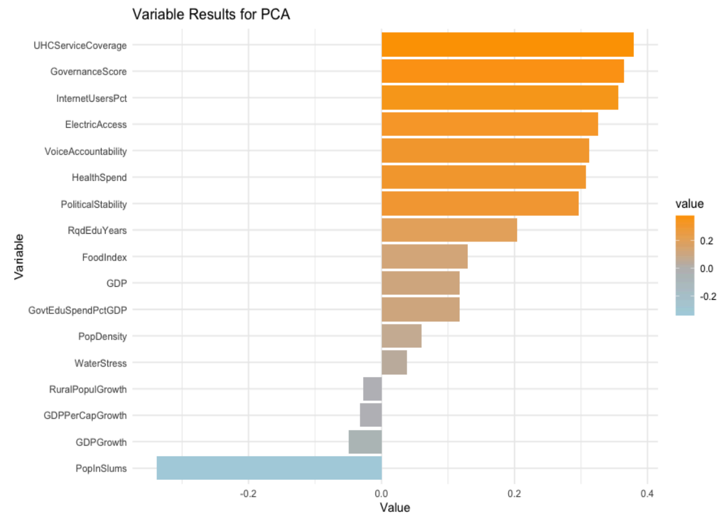


Figure 11

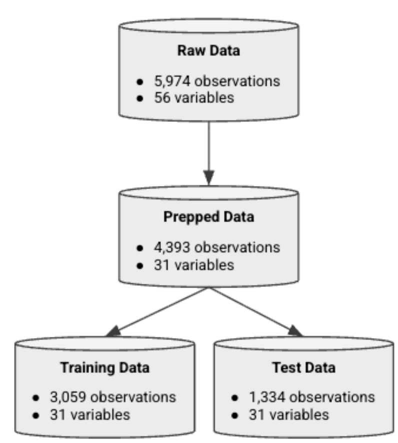


An more important finding from the PCA exercise is that the first Principal Component (PC1) shows a 92.5% correlation with our HDI response variable - and, as Figure 11 shows, Internet Usage is the third most heavily weighted factor in PC1, which bodes well for our models proving out our hypothesis regarding the connection between internet access and HDI.   While this proved illuminating, we subsequently chose to EXCLUDE internet access from the Principal Component modeling when we produced our final models, in order to isolate the unique impact of internet access versus all the other potential predictors in any models built using our Principal Components.

Before And After Data Sets

As shown in Figure 12, the results of the data cleaning and preprocessing left us with training and test data sets with more than ample data to support robust predictive modeling with defensible results

Figure 12
Results of Cleaning and Preprocessing



Stage 2: Algorithm Selection

For our study, we built and evaluated models using a number of different algorithms, in 4 categories:

- *Regression (linear and logistic)*

    - These algorithms, which are the simplest to explain, assume linearity across the variables. They included:
        - OLS regression to predict numeric HDI measures
        - Logistic regression to predict the HDI Category
        - "Panel regression" model with fixed effects, to predict the HDI index using 3-year lagged change in internet access, with fixed effects of year and country.
        - We did not account for "overfitting" in these models. These models were intentionally simplistic to pave the way for more complex models which will address collinearity

<u>Why the "Panel Regression"</u>

- Time series panel data suffers from "autocorrelation" in the data - a simple analysis can show that 99% of a given country/year's HDI index variance could be explained by the previous year's index!
- Strong linear correlation <> causality - one could as easily argue "reverse causality" - that countries with higher HDI have higher internet access because more people can afford it - a concept in econometrics referred to as "endogeneity".
- The panel regression mitigates autocorrelation, while using 3 year lagged change in Internet Access as the predictor more strongly suggests causation

- *Machine learning: Tree-based algorithms*
    - In order to improve the ability to predict HDI while isolating the impact of internet access, multivariate algorithms that consider a broader set of predictors were required.
    - While we reduced collinearity in our data through the use of binning, multi-variate regression algorithms (e.g. Ridge/ Lasso) would likely still suffer from collinearity.
    - After experimentation, we chose to build a number of tree-based models using the XG-Boost algorithm, since the ability to assess "feature importance" within these models provides a greater degree of interpretability than more complex "black box" machine learning models, and would allow us to assess the impact of Internet Access.
    - Mean Average Error ("MAE") was used as the prediction error measure for quantitative predictions of the HDI_Index, and prediction accuracy was used for HDI Category.

- We tuned and controlled for overfitting by using a hyperparameter tuning grid and k-fold cross-validation as follows:
  - Tuning Grid:
    - Learning Rate (eta) : 0.01, 0.1 and 0.3
    - Maximum Tree Depth (max_depth): 6, 12 and 18
    - Subsamples: 0.6, 0.8 and 1.0
  - Validation
    - 5 folds
    - 70/30 split, keeping countries intact
    - Model with best average error across folds selected
- The assumptions in these algorithms, and how they were evaluated, include:
  - *No major class imbalance:* The majority class for our binned HDI index response variable contains only 28% of the observations
  - *Categorical variables are factored properly:* Income Group and HDI Category are both designed as ordered factors
  - *Data is independent (no leakage):* We addressed some of this by splitting the test vs training data by country as described above; However - this is a likely weakness in our modeling efforts due to autocorrelation - whereby a time-period's data is heavily influenced, and could be predicted by, previous time-periods' data.
  - *Sufficient trees will split the classes:* We varied the number of trees to ensure this

- *Machine learning: Neural Networks*
  - To try to improve on our model predictions, we used a series of feed-forward neural network algorithms seeking to predict the HDI Index using all the predictors except the Principal Components.
  - MAE was used as the error rate measure
  - We tuned and controlled for overfitting using a tuning grid with validation:
    - Tuning grid:
      - Epochs: 1,000 and 2,000
      - Learning rate: 0.1, 0.01, 0.001 and 0.0001
      - Batch sizes: 16, 32, 64 and 128
      - Optimizers: ReLU, Sigmoid and Tanh
    - Validation:

- 10% validation set was held out of the training data
- Early stopping patience of 50 epochs with no change in validation error

Stage 3: Modeling

While we built and evaluated a quite large number of models, the following is a sampling of the most stable and interpretable models produced:

1) *Models to predict the Quantitative HDI Index*

- To compare model results, we calculated RMSE, MAE, MAPE and Median AE for all models based on how they performed on held-out test data.
- Model summary results are in Figures 12 and 13.

Figure 13: Predicting the HDI Index

#	Algorithm	Predictors	RMSE	MAE	MAPE	Median AE	Comments
1q	Simple OLS	InternetUsersPct	0.0955	0.0770	12.2%	0.0661	$R^2 = 0.634$
2q	Quadratic OLS	InternetUsersPct	0.0954	0.0769	12.2%	0.0652	$R^2 = 0.635$
3q	Simple OLS	Cumulative3YrChg_ InternetUsersPct	0.148	0.120	18.8%	0.1049	$R^2 = 0.121$
4q[5]	Panel Regression	Cumulative3YrChg_ InternetUsersPct	NA	NA	NA	NA	$R^2 = 0.059$
5q	XGBoost	All except Principal Components	0.0474	0.0352	5.9%	0.0260	$R^2 = 0.918$ Training RMSE = 0.042
6q[6]	XGBoost	InternetUsersPct & PC1-PC5	N/A	N/A	N/A	N/A	N/A
7q	Neural Network[7]	All except Principal Components	N/A	0.1352	N/A	N/A	N/A

Observations:

- Simple OLS has the most explainability but only shows correlation

[5] This model was not used to run predictions; it was simply used as a starting point to illustrate the use of panel regression and better quantify the relationship between internet access and HDI
[6] This model was not completed in time for the publishing of this study
[7] Because the stability of this model was so poor and the MAE result inferior to other models, predictions using held-out test data were not run. Reported MAE is from training data

- Using 3 year change in HDI in simple regression gets us closer to "truth" regarding the lagged impact of investment in internet infrastructure
- The panel regression simply attempts to isolate the impact of lagged internet access on HDI - not to predict HDI per se
- The XGBoost regression model achieved strong predictive performance, with an R^2 of 0.92 and a test RMSE of 0.0474 (vs. 0.042 training RMSE) on an HDI scale of 0–1, indicating excellent fit and virtually no overfitting (See Appendix C)
- Neural networks performed poorly and were abandoned (See Appendix D)

Figure 14: Predicting the Change in HDI

#	Algorithm	Predictors	RMSE	MAE	MAPE	Median AE	Comments
8q	Simple OLS	YearlyChgInternet	0.0067	0.0042	95.1%	0.0030	$R^2 = 0.000$
9q	Simple OLS	Cumulative3YrChg_ InternetUsersPct	0.0126	0.00944	131.6%	0.0075	$R^2 = 0.002$

Observation: These models performed so poorly that this approach is unsupportable

2) *Models to predict HDI Category*

- To compare model results, we calculated Log Loss and Accuracy[8] for all models based on how they performed on held-out test data.
- Model summary results are in Figure 14

[8]Since the majority class only has 28% of the observations, we determined that given the balanced classes, accuracy would suffice for performance measurement and comparison.

#	Algorithm	Predictors	Tuning / Validation	Log Loss	Accuracy	Comments
1c	Logistic Regression	InternetUsersPct	None	0.924	56.4%	
2c	XGBoost	All except Principal Components	Tuning Grid Loop/ K-folds CV	0.600	76.6%	• 77.4% accuracy on training data; • 0.616 logloss on training data
3c[9]	XGBoost	InternetUsersPct & PC1-PC5	Tuning Grid Loop/ K-folds CV	N/A	N/A	N/A

Figure 15: Predicting HDI Category

Observations:

- The logistic regression model is simply not accurate enough, again demonstrating the internet access alone does not predict HDI (even at a category level)
- XGBoost model accuracy is much stronger, and generalized extremely well to the test set, with both accuracy and mean Log Loss very much in line (See Appendix E)

Final Models

We believe that a combination of three models best supports our conclusions:

1) **Model 1q (Simple Two-Variable OLS Regression analysis)**

   This model expresses HDI as a function of Internet access using the following equation:

   $HDI = 0.0394 * InternetUsersPCT + 0.467 + \varepsilon$

   This suggests that each 1% difference in Internet Access equates to a 0.0394 difference in HDI. After applying the model to our test data, the model does establish a simple base of support for our hypothesis, as it shows that 63% of the variability in HDI can be explained via Internet Access. While this does not show causality, this is a strong baseline of support for correlation between internet access and HDI.   The model's statistics are shown in Figure 15

9 This model was not completed in time for the publishing of this study

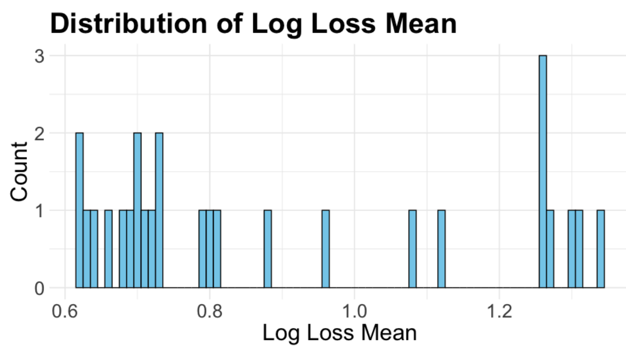
	r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.661	0.660	0.0932	5524.	0	1	2712.	-5419.	-5401.	24.6

Figure 16: Linear Regression Model Reults

2) **Model 2c (XGBoost model predicting HDI Category)**

While both XGBoost models perform well, we believe predicting HDI by category is more explainable. Model 2c, which was the one with lowest mean log loss (≈0.60) across folds (See log loss mean distribution in Figure 16 and Appendix E), shows solid prediction accuracy at

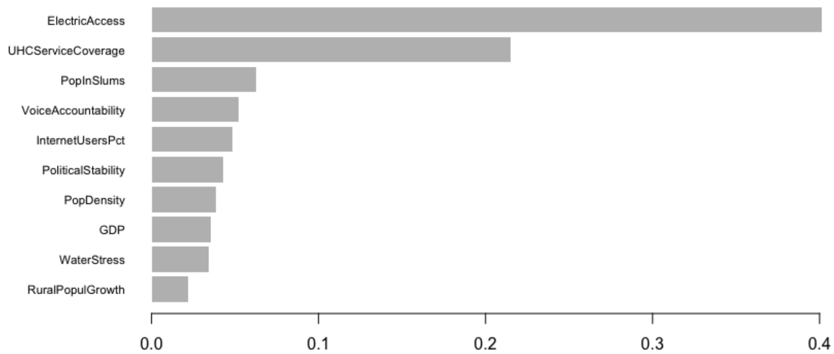
Figure 17



77%, with the confusion matrix in Figure 17 demonstrating that "errors" are never more than one class away from truth. In addition, a review of the feature importance matrix shows that InternetUsersPct is the fifth most important predictor HDI in this model, lending further support to our hypothesis.

Figure 18: Feature Importance Matrix for Predicting HDI Category

	Predicted			
Actual	0	1	2	3
0	202	68	0	0
1	33	120	56	0
2	0	17	197	11
3	0	0	73	326



Finally - this model generalized quite well to unseen data, with both prediction accuracy and log loss mean on the held-out test data very much in line with the training model results.

3) **Model 4q (Panel Regression Predicting HDI)**

This model predicts HDI (while fixing year and country to minimize autocorrelation) using 3 year lagged change in Internet Access. The model shows a quantifiable positive relationship between the two, as shown in Figure 18

Figure 19: Panel Regression Model Reults

```
Coefficients:
                                  Estimate Std. Error t-value  Pr(>|t|)
Cumulative3yrChg_InternetUsersPct 5.9183e-04 4.6006e-05  12.864 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Total Sum of Squares:    0.78089
Residual Sum of Squares: 0.73455
R-Squared:      0.059348
Adj. R-Squared: 0.0044792
F-statistic: 165.49 on 1 and 2623 DF, p-value: < 2.22e-16
```

This suggests that a 10% change in Internet Access over a 3 year period will directly lead to a 0.005 change in HDI, which is statistically significant given that HDI is on a scale of 0 to 1, with most values above 0.4   While this model needs significant further refinement, the minimization of autocorrelation and the use of a lagged predictor present the strongest support we could find for a positive, quantitative, causative relationship between growth in internet access and HDI, directly supporting our hypothesis.

Conclusions and Recommendations

- Our study shows that there is a strong correlation between growth in digital inclusion and improved population wellbeing as measured by HDI.
- The study does NOT support predicting wellbeing based on digital inclusion alone, but it does demonstrate growth in internet access has positive predictive value and correlation to HDI categories as defined by the United Nations, since it is an important variable in a model that can predict HDI categories with meaningful accuracy.
- The study also lays the groundwork for a more quantifiable method for predicting how much growth in HDI can be expected from sustained investment in internet access alone .
- The study shows the ongoing importance of expanding access to **electricity** to both drive HDI, and to facilitate the HDI accelerant effects of internet access. Having reliable access to sources of electricity dominates the ability of new users to engage in "meaningful connectivity," but there are deeper issues to navigate through infrastructure initiatives and their funding sources that can also improve well-being of these individuals (Leiva Vilaplana et al., 2025).
- Our models reinforce the importance of health care to HDI, and research suggests a scaling effect may be present if investments in electricity and internet access are made:
  - Currently the health outcomes of individuals in underdeveloped nations are greatly impacted by levels of pollution introduced by the production of electricity via more traditional means. Moving towards more renewable energy sources with funding from private-public partnerships that produce less pollution is a current trend in improving the lives of those in underprivileged and rural communities (Song et al., 2022).

    - The health outcomes of individuals in G7 nations have improved in part because internet users can access health providers remotely for earlier interventions in chronic conditions, and through the ability to access health-related information online that otherwise would not be available to them. While there are concerns about the negative impacts of access to the internet and social media due to misinformation and mental health impacts, these areas do not outweigh the benefits of having access to more advanced healthcare options (Kaya et al., 2025).
    - We believe this study conclusively enough to support a prescriptive recommendation: **Technology companies and e-commerce companies seeking to benefit from the "Next Billion Users", and aid organizations and local governments seeking to improve population wellbeing, should partner together to invest in expanding internet access in underserved areas, including investing in renewable energy sources to expand access to electricity where necessary, in order to achieve their mutual goals.**

Next Steps

This study could be improved in the following ways:

- Re-running the XGBoost models using internet access as a predictor along with the top 5 principal components (redesigned to exclude internet access)
- "Bin" electricity access into [Low/Medium/High] bins, and build separate models for each subset of data, removing Electricity Access as a predictor and instead controlling for it, to better isolate the impact of internet access as a separate factor in HDI growth
- "Bin" countries by Income Group as suggested in EDA, and build separate models for each subset of data, to better isolate the differential impact of internet access for each cohort

- More fully explore lagged features to attempt to pinpoint how much time it takes for investment in internet access to flow through to HDI
- More fully explore panel regression to model the direct impact of internet access on HDI

Code Availability

Our analysis was performed using R version 4.5.1 (2025-06-13) and Python 3.13.5 with the following key libraries: Pandas, Numpy, Scikit-learn, and Matplotlib (add R libraries) with the full list of additional libraries and packages listed in the requirements.txt file.

The code used for data processing, analysis, and visualization is publicly available in this repository: https://github.com/Merrimack-Capstone/MSDS-capstone. The Read.me file outlines the code modules and what components of the data extraction, processing, visualization and analysis each performs

To reproduce our results, you'll need to install the dependencies listed in the requirements.txt file. The raw data backup files used in this project are available in the Data subfolder in the Github repo.

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Appendix A

Data Dictionary

Short Name	Category / "Topic"	Long Name	Type	Source
HDI_Category	N/A	N/A	Response	UNDP
Lag1_InternetUsersPct	N/A	N/A	Predictor	World Bank (Data)
Lag2_InternetUsersPct	N/A	N/A	Predictor	World Bank (Data)
YearlyChgInternet	N/A	N/A	Predictor	World Bank (Data)
Lag1_YearlyChgInternet	N/A	N/A	Predictor	World Bank (Data)
Lag2_YearlyChgInternet	N/A	N/A	Predictor	World Bank (Data)
GovernanceScore	N/A	N/A	Predictor	World Bank (Data)
HealthSpend	N/A	N/A	Predictor	World Bank (Data)
seriesCode[1]	N/A	N/A	Predictor	ITU - Mobile
entityIso	N/A	N/A	Predictor	ITU - Mobile
dataValue	N/A	N/A	Predictor	ITU - Mobile
dataYear	N/A	N/A	Predictor	ITU - Mobile

[1]

Short Name	Category / “Topic”	Long Name	Type	Source
seriesCode[2]	N/A	N/A	Predictor	ITU - Fixed
entityIso	N/A	N/A	Predictor	ITU - Fixed
dataValue	N/A	N/A	Predictor	ITU - Fixed
dataYear	N/A	N/A	Predictor	ITU - Fixed
Country	N/A	N/A	Response	UNDP
Year[3]	N/A	N/A	Response	UNDP
HDI Index	N/A	N/A	Response	UNDP
Gini Coefficient	N/A	N/A	Response	UNDP
Country Name	N/A	N/A	Identifier	World Bank (Data)
Country Code	N/A	N/A	Identifier	World Bank (Data)
Income Level	N/A	N/A	Predictor / Control	World Bank (Country)
Year[5]	N/A	N/A	Identifier	World Bank (Data)
SE.SEC.CUAT.PO.ZS	Education: Outcomes	Educational attainment, at least completed post-secondary, population 25+, total (%) (cumulative)	Response	World Bank (Data)
SE.SEC.CUAT.UP.ZS	Education: Outcomes	Educational attainment, at least completed upper secondary, population 25+, total (%) (cumulative)	Response	World Bank (Data)
SE.TER.CUAT.BA.ZS	Education: Outcomes	Educational attainment, at least Bachelor's or equivalent, population 25+, total (%) (cumulative)	Response	World Bank (Data)

2”

3

Short Name	Category / "Topic"	Long Name	Type	Source
SE.TER.CUAT.DO.ZS	Education: Outcomes	Educational attainment, Doctoral or equivalent, population 25+, total (%) (cumulative)	Response	World Bank (Data)
SE.TER.CUAT.MS.ZS	Education: Outcomes	Educational attainment, at least Master's or equivalent, population 25+, total (%) (cumulative)	Response	World Bank (Data)
SE.TER.CUAT.ST.ZS	Education: Outcomes	Educational attainment, at least completed short-cycle tertiary, population 25+, total (%) (cumulative)	Response	World Bank (Data)
AG.PRD.FOOD.XD	Environment: Agricultural production	Food production index (2014-2016 = 100)	Predictor	World Bank (Data)
EN.POP.DNST	Environment: Density & urbanization	Population density (people per sq. km of land area)	Predictor	World Bank (Data)
EN.POP.SLUM.UR.ZS	Environment: Density & urbanization	Population living in slums (% of urban population)	Predictor	World Bank (Data)
SP.RUR.TOTL.ZG	Environment: Density & urbanization	Rural population growth (annual %)	Predictor	World Bank (Data)
EG.ELC.ACCS.ZS	Environment: Energy production & use	Access to electricity (% of population)	Predictor	World Bank (Data)
ER.H2O.FWST.ZS	Environment: Freshwater	Level of water stress: freshwater withdrawal as a proportion of available freshwater resources	Predictor	World Bank (Data)
FX.OWN.TOTL.ZS	Financial Sector: Access	Account ownership at a financial institution or with a mobile-money-service provider (% of population ages 15+)	Predictor	World Bank (Data)
SN.ITK.MSFI.ZS	Health	Prevalence of moderate or severe food insecurity in the population (%)	Predictor	World Bank (Data)
SH.XPD.CHEX.PC.CD	Health: Health systems	Current health expenditure per capita (current US$)	Predictor	World Bank (Data)

Short Name	Category / “Topic”	Long Name	Type	Source
SH.XPD.GHED.PC.CD	Health: Health systems	Domestic general government health expenditure per capita (current US$)	Predictor	World Bank (Data)
SH.STA.WASH.P5	Health: Mortality	Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (per 100,000 population)	Predictor	World Bank (Data)
SP.DYN.LE00.IN	Health: Mortality	Life expectancy at birth, total (years)	Response	World Bank (Data)
SH.UHC.SRVS.CV.XD	Health: Universal Health Coverage	UHC service coverage index	Predictor	World Bank (Data)
IT.NET.BBND.P2	Infrastructure: Communications	Fixed broadband subscriptions (per 100 people)	Predictor	World Bank (Data)
IT.NET.USER.ZS	Infrastructure: Communications	Individuals using the Internet (% of population)	Predictor	World Bank (Data)
GB.XPD.RSDV.GD.ZS	Infrastructure: Technology	Research and development expenditure (% of GDP)	Predictor	World Bank (Data)
IS.SHP.GCNW.XQ	Infrastructure: Transportation	Liner shipping connectivity index (maximum value in 2004 = 100)	Predictor	World Bank (Data)
SI.POV.GINI	Poverty: Income distribution	Gini index	Response	World Bank (Data)
VC.IHR.PSRC.P5	Public Sector: Conflict & fragility	Intentional homicides (per 100,000 people)	Predictor	World Bank (Data)
CC.EST	Public Sector: Policy & institutions	Control of Corruption: Estimate	Predictor	World Bank (Data)
GE.EST	Public Sector: Policy & institutions	Government Effectiveness: Estimate	Predictor	World Bank (Data)

Short Name	Category / "Topic"	Long Name	Type	Source
PV.EST	Public Sector: Policy & institutions	Political Stability and Absence of Violence/Terrorism: Estimate	Predictor	World Bank (Data)
RL.EST	Public Sector: Policy & institutions	Rule of Law: Estimate	Predictor	World Bank (Data)
VA.EST	Public Sector: Policy & institutions	Voice and Accountability: Estimate	Predictor	World Bank (Data)
SM.POP.TOTL.ZS	Social Protection & Labor: Migration	International migrant stock (% of population)	Predictor	World Bank (Data)

Appendix B

Exploratory Data Analysis

Tables, Visualizations, and Interpretations

Table 1: Summary Statistics (Variables 1 to 8)

CountryCode	CountryName	year	HDI_Index	IHDI_Index	GiniCoeff	FoodIndex	CorruptionScore
Length:5974	Length:5974	Min. :1995	Min. :0.2380	Min. :0.1340	Min. : 4.819	Min. : 17.00	Min. :-1.96956
Class :character	Class :character	1st Qu.:2002	1st Qu.:0.5700	1st Qu.:0.3990	1st Qu.:11.060	1st Qu.: 76.58	1st Qu.:-0.82041
Mode :character	Mode :character	Median :2009	Median :0.7110	Median :0.5930	Median :19.183	Median : 94.52	Median :-0.29892
NA	NA	Mean :2009	Mean :0.6899	Mean :0.5783	Mean :21.317	Mean : 91.33	Mean :-0.05974
NA	NA	3rd Qu.:2016	3rd Qu.:0.8140	3rd Qu.:0.7500	3rd Qu.:31.878	3rd Qu.:102.80	3rd Qu.: 0.59246
NA	NA	Max. :2023	Max. :0.9720	Max. :0.9230	Max. :51.891	Max. :578.71	Max. : 2.45912
NA	NA	NA	NA's :422	NA's :3677	NA's :3670	NA's :720	NA's :1204

Table 2: Summary Statistics (Variables 9 to 16)

ElectricAccess	PopDensity	PopInSlums	WaterStress	BankingAccess	RnDInvest	GovtEffectiveness	ShippingIndex
Min. : 0.80	Min. : 1.438	Min. : 0.00	Min. : 0.0258	Min. : 0.40	Min. :0.0054	Min. :-2.44023	Min. : 0.6032
1st Qu.: 63.20	1st Qu.: 30.875	1st Qu.: 1.55	1st Qu.: 3.7148	1st Qu.: 31.07	1st Qu.:0.2383	1st Qu.:-0.78651	1st Qu.: 7.5392
Median : 98.10	Median : 77.922	Median :21.18	Median : 11.1751	Median : 54.05	Median :0.6029	Median :-0.20432	Median : 15.1677
Mean : 79.23	Mean : 298.200	Mean :28.73	Mean : 65.0840	Mean : 57.04	Mean :0.9675	Mean :-0.06367	Mean : 25.1422
3rd Qu.:100.00	3rd Qu.: 176.610	3rd Qu.:52.80	3rd Qu.: 39.6974	3rd Qu.: 86.76	3rd Qu.:1.3904	3rd Qu.: 0.59433	3rd Qu.: 34.1564
Max. :100.00	Max. :18822.891	Max. :99.81	Max. :3850.5000	Max. :100.00	Max. :6.0192	Max. : 2.46966	Max. :171.1775
NA's :627	NA's :583	NA's :4011	NA's :1505	NA's :5412	NA's :3666	NA's :1223	NA's :3592

Table 3: Summary Statistics (Variables 17 to 24)

FixedIntSubs	InternetUsersPct	PoliticalStability	RuleOfLaw	RqdEduYears	GovtEduSpendPctGDP	MortalityFromDirtyness	UHCServiceCoverage
Min. : 0.0000	Min. : 0.00	Min. :-3.31295	Min. :-2.59088	Min. : 4.00	Min. : 0.000	Min. : 0.40	Min. :11.00
1st Qu.: 0.2673	1st Qu.: 2.81	1st Qu.:-0.69566	1st Qu.:-0.82174	1st Qu.: 8.00	1st Qu.: 3.139	1st Qu.: 2.95	1st Qu.:44.00
Median : 3.8683	Median : 21.00	Median : 0.02612	Median :-0.20532	Median : 9.00	Median : 4.232	Median : 6.10	Median :63.00
Mean :10.5358	Mean : 32.58	Mean :-0.06279	Mean :-0.06373	Mean : 9.42	Mean : 4.418	Mean : 16.96	Mean :59.39
3rd Qu.:18.4017	3rd Qu.: 61.40	3rd Qu.: 0.79477	3rd Qu.: 0.69307	3rd Qu.:11.00	3rd Qu.: 5.428	3rd Qu.: 25.55	3rd Qu.:75.00
Max. :62.2147	Max. :100.00	Max. : 1.75868	Max. : 2.12476	Max. :17.00	Max. :15.863	Max. :108.10	Max. :91.00
NA's :2041	NA's :675	NA's :1160	NA's :1114	NA's :1432	NA's :2288	NA's :5791	NA's :4630

Table 4: Summary Statistics (Variables 25 to 32)

HealthSpendPerCapita	GovtHealthSpendPerCapita	MigrantsPct	FoodInsecurityPct	RuralPopulGrowth	VoiceAccountability	Homicides	Income Group
Min. : 4.176	Min. : 0.1489	Min. : 0.036	Min. : 2.00	Min. :-235.9259	Min. :-2.31340	Min. : 0.000	Length:5974
1st Qu.: 63.631	1st Qu.: 21.9384	1st Qu.: 1.212	1st Qu.: 8.90	1st Qu.: -0.5789	1st Qu.:-0.88313	1st Qu.: 1.228	Class :character
Median : 261.295	Median : 128.2416	Median : 3.663	Median :22.30	Median : 0.4574	Median :-0.00619	Median : 2.861	Mode :character
Mean : 959.978	Mean : 648.7002	Mean : 8.791	Mean :29.09	Mean : 0.3933	Mean :-0.04756	Mean : 7.939	NA
3rd Qu.: 890.483	3rd Qu.: 557.3557	3rd Qu.:10.632	3rd Qu.:45.30	3rd Qu.: 1.6167	3rd Qu.: 0.85004	3rd Qu.: 8.879	NA
Max. :12434.434	Max. :7871.2450	Max. :88.404	Max. :89.20	Max. : 15.0306	Max. : 1.80099	Max. :138.774	NA
NA's :1617	NA's :1630	NA's :5009	NA's :4953	NA's :458	NA's :1112	NA's :2755	NA

COMMENTS:

- Basic summary of every variable, showing type, distribution, mean/median, and missingness
- Can use this to start understanding our data better and to deal with sparseness
- Pretty dense, but we can already see a few things:
  - We have 400+ observations without our core HDI_Index response
  - We'll need to reconsider whether to use IHDI given over 60% is missing
  - There are other fields with large % missing (too sparse)

Table 5: Data summary

Name	FirstCutData
Number of rows	5974
Number of columns	32
Column type frequency:	
character	3
numeric	29
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
CountryCode	0	1.00	3	9	0	206	0
CountryName	319	0.95	4	38	0	195	0
Income Group	377	0.94	10	19	0	4	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
year	0	1.00	2009.00	8.37	1995.00	2002.00	2009.00	2016.00	2023.00	
HDI_Index	422	0.93	0.69	0.16	0.24	0.57	0.71	0.81	0.97	
IHDI_Index	3677	0.38	0.58	0.20	0.13	0.40	0.59	0.75	0.92	
GiniCoeff	3670	0.39	21.32	11.35	4.82	11.06	19.18	31.88	51.89	
FoodIndex	720	0.88	91.33	28.06	17.00	76.57	94.52	102.80	578.71	
CorruptionScore	1204	0.80	-0.06	1.00	-1.97	-0.82	-0.30	0.59	2.46	
ElectricAccess	627	0.90	79.23	30.10	0.80	63.20	98.10	100.00	100.00	
PopDensity	583	0.90	298.20	1389.80	1.44	30.87	77.92	176.61	18822.89	
PopInSlums	4011	0.33	28.73	27.32	0.00	1.55	21.18	52.80	99.81	
WaterStress	1505	0.75	65.08	266.68	0.03	3.71	11.18	39.70	3850.50	
BankingAccess	5412	0.09	57.04	30.13	0.40	31.06	54.05	86.76	100.00	
RnDInvest	3666	0.39	0.97	0.98	0.01	0.24	0.60	1.39	6.02	
GovtEffectiveness	1223	0.80	-0.06	1.00	-2.44	-0.79	-0.20	0.59	2.47	
ShippingIndex	3592	0.40	25.14	25.22	0.60	7.54	15.17	34.16	171.18	

COMMENTS:

- Similar to the Summary data above
- Adds a “complete_rate” variable which shows what % of the data is populated, which is super helpful

Table 6: Correlation Matrix: Columns 1 to 8

	HDI_Index	IHDI_Index	GiniCoeff	FoodIndex	CorruptionScore	ElectricAccess	PopDensity	PopInSlums
HDI_Index	1.00	0.98	-0.92	0.29	0.71	0.85	0.17	-0.82
IHDI_Index	0.98	1.00	-0.97	-0.10	0.72	0.80	0.14	-0.85
GiniCoeff	-0.92	-0.97	1.00	0.10	-0.62	-0.79	-0.09	0.84
FoodIndex	0.29	-0.10	0.10	1.00	0.13	0.20	0.08	-0.20
CorruptionScore	0.71	0.72	-0.62	0.13	1.00	0.47	0.20	-0.57
ElectricAccess	0.85	0.80	-0.79	0.20	0.47	1.00	0.10	-0.75
PopDensity	0.17	0.14	-0.09	0.08	0.20	0.10	1.00	-0.11
PopInSlums	-0.82	-0.85	0.84	-0.20	-0.57	-0.75	-0.11	1.00
WaterStress	0.12	0.08	-0.08	0.01	0.04	0.13	0.05	-0.14
BankingAccess	0.82	0.81	-0.72	0.04	0.75	0.56	0.18	-0.69
RnDInvest	0.63	0.62	-0.47	0.12	0.69	0.27	0.07	-0.47
GovtEffectiveness	0.79	0.83	-0.74	0.12	0.92	0.57	0.22	-0.64
ShippingIndex	0.46	0.46	-0.38	0.10	0.35	0.34	0.34	-0.36
FixedIntSubs	0.73	0.88	-0.80	0.14	0.64	0.47	0.24	-0.57
InternetUsersPct	0.72	0.87	-0.82	0.29	0.56	0.55	0.15	-0.64
PoliticalStability	0.59	0.63	-0.58	0.10	0.75	0.41	0.14	-0.56
RuleOfLaw	0.74	0.77	-0.68	0.13	0.94	0.50	0.18	-0.63
RqdEduYears	0.41	0.39	-0.38	0.13	0.26	0.42	-0.01	-0.36
GovtEduSpendPctGDP	0.26	0.28	-0.28	0.11	0.35	0.20	-0.15	-0.33
MortalityFromDirtyness	-0.80	-0.79	0.80	0.15	-0.42	-0.82	-0.07	1.00
UHCServiceCoverage	0.92	0.92	-0.86	0.25	0.61	0.81	0.13	-0.79
HealthSpendPerCapita	0.64	0.68	-0.57	0.09	0.73	0.36	0.26	-0.48
GovtHealthSpendPerCapita	0.63	0.67	-0.57	0.09	0.72	0.35	0.29	-0.47
MigrantsPct	0.41	0.50	-0.41	0.06	0.43	0.34	0.37	-0.38
FoodInsecurityPct	-0.83	-0.85	0.84	0.14	-0.62	-0.78	-0.13	0.77
RuralPopulGrowth	-0.24	-0.52	0.50	-0.02	-0.30	-0.23	-0.03	0.48
VoiceAccountability	0.59	0.65	-0.58	0.10	0.77	0.40	0.08	-0.55
Homicides	-0.24	-0.30	0.26	-0.01	-0.24	-0.06	-0.09	0.17

Table 7: Correlation Matrix: Columns 9 to 16

	WaterStress	BankingAccess	RnDInvest	GovtEffectiveness	ShippingIndex	FixedIntSubs	InternetUsersPct	PoliticalStability
HDI_Index	0.12	0.82	0.63	0.79	0.46	0.73	0.72	0.59
IHDI_Index	0.08	0.81	0.62	0.83	0.46	0.88	0.87	0.63
GiniCoeff	-0.08	-0.72	-0.47	-0.74	-0.38	-0.80	-0.82	-0.58
FoodIndex	0.01	0.04	0.12	0.12	0.10	0.14	0.29	0.10
CorruptionScore	0.04	0.75	0.69	0.92	0.35	0.64	0.56	0.75
ElectricAccess	0.13	0.56	0.27	0.57	0.34	0.47	0.55	0.41
PopDensity	0.05	0.18	0.07	0.22	0.34	0.24	0.15	0.14
PopInSlums	-0.14	-0.69	-0.47	-0.64	-0.36	-0.57	-0.64	-0.56
WaterStress	1.00	0.08	-0.08	0.04	0.05	-0.04	0.10	0.03
BankingAccess	0.08	1.00	0.66	0.80	0.49	0.81	0.81	0.65
RnDInvest	-0.08	0.66	1.00	0.72	0.46	0.59	0.49	0.43
GovtEffectiveness	0.04	0.80	0.72	1.00	0.49	0.69	0.62	0.71
ShippingIndex	0.05	0.49	0.46	0.49	1.00	0.49	0.46	0.09
FixedIntSubs	-0.04	0.81	0.59	0.69	0.49	1.00	0.83	0.46
InternetUsersPct	0.10	0.81	0.49	0.62	0.46	0.83	1.00	0.41
PoliticalStability	0.03	0.65	0.43	0.71	0.09	0.46	0.41	1.00
RuleOfLaw	0.05	0.78	0.70	0.94	0.39	0.65	0.57	0.78
RqdEduYears	-0.02	0.23	0.11	0.28	0.16	0.32	0.38	0.20
GovtEduSpendPctGDP	0.06	0.36	0.32	0.27	-0.02	0.12	0.17	0.33
MortalityFromDirtyness	-0.12	NA	-0.23	-0.57	-0.27	-0.56	-0.76	-0.45
UHCServiceCoverage	0.10	0.76	0.54	0.70	0.46	0.73	0.82	0.51
HealthSpendPerCapita	0.01	0.66	0.71	0.72	0.43	0.75	0.64	0.46
GovtHealthSpendPerCapita	0.02	0.66	0.69	0.71	0.39	0.74	0.63	0.46
MigrantsPct	0.61	NA	0.21	0.43	0.17	0.35	0.37	0.36
FoodInsecurityPct	-0.08	-0.69	-0.49	-0.74	-0.47	-0.72	-0.77	-0.51
RuralPopulGrowth	-0.13	-0.44	-0.05	-0.31	-0.24	-0.12	-0.16	-0.31
VoiceAccountability	-0.14	0.65	0.54	0.75	0.13	0.54	0.44	0.68
Homicides	-0.09	-0.28	-0.31	-0.24	-0.23	-0.22	-0.20	-0.18

COMMENTS:

- Tables 6 through 9 show the correlation between all the features, from -1 (perfect negative correlation) to +1 (perfect positive correlation)
- The rules of thumb we are using are:
    - If absolute value of correlation is 0.9 or greater, one of those features will be dropped from statistical predictive models due to collinearity
    - If absolute value of correlation is between 0.7 and 0.9, we will evaluate collinearity impact (e.g. using VIF)
- This data is easier to deal with visually, which can be found on subsequent pages of this report

Table 8: Correlation Matrix: Columns 17 to 24

	RuleOfLaw	RqdEduYears	GovtEduSpendPctGDP	MortalityFromDirtyness	UHCServiceCoverage	HealthSpendPerCapita	GovtHealthSpendPerCapita	MigrantsPct
HDI_Index	0.74	0.41	0.26	-0.80	0.92	0.64	0.63	0.41
IHDI_Index	0.77	0.39	0.28	-0.79	0.92	0.68	0.67	0.50
GiniCoeff	-0.68	-0.38	-0.28	0.80	-0.86	-0.57	-0.57	-0.41
FoodIndex	0.13	0.13	0.11	0.15	0.25	0.09	0.09	0.06
CorruptionScore	0.94	0.26	0.35	-0.42	0.61	0.73	0.72	0.43
ElectricAccess	0.50	0.42	0.20	-0.82	0.81	0.36	0.35	0.34
PopDensity	0.18	-0.01	-0.15	-0.07	0.13	0.26	0.29	0.37
PopInSlums	-0.63	-0.36	-0.33	1.00	-0.79	-0.48	-0.47	-0.38
WaterStress	0.05	-0.02	0.06	-0.12	0.10	0.01	0.02	0.61
BankingAccess	0.78	0.23	0.36	NA	0.76	0.66	0.66	NA
RnDInvest	0.70	0.11	0.32	-0.23	0.54	0.71	0.69	0.21
GovtEffectiveness	0.94	0.28	0.27	-0.57	0.70	0.72	0.71	0.43
ShippingIndex	0.39	0.16	-0.02	-0.27	0.46	0.43	0.39	0.17
FixedIntSubs	0.65	0.32	0.12	-0.56	0.73	0.75	0.74	0.35
InternetUsersPct	0.57	0.38	0.17	-0.76	0.82	0.64	0.63	0.37
PoliticalStability	0.78	0.20	0.33	-0.45	0.51	0.46	0.46	0.36
RuleOfLaw	1.00	0.26	0.29	-0.47	0.62	0.71	0.70	0.42
RqdEduYears	0.26	1.00	0.16	-0.35	0.45	0.28	0.26	0.11
GovtEduSpendPctGDP	0.29	0.16	1.00	-0.13	0.25	0.20	0.21	-0.01
MortalityFromDirtyness	-0.47	-0.35	-0.13	1.00	-0.80	-0.35	-0.34	NA
UHCServiceCoverage	0.62	0.45	0.25	-0.80	1.00	0.58	0.56	0.35
HealthSpendPerCapita	0.71	0.28	0.20	-0.35	0.58	1.00	0.97	0.38
GovtHealthSpendPerCapita	0.70	0.26	0.21	-0.34	0.56	0.97	1.00	0.37
MigrantsPct	0.42	0.11	-0.01	NA	0.35	0.38	0.37	1.00
FoodInsecurityPct	-0.68	-0.30	-0.22	0.70	-0.79	-0.52	-0.53	-0.39
RuralPopulGrowth	-0.31	-0.12	-0.08	0.49	-0.46	-0.10	-0.11	-0.05
VoiceAccountability	0.82	0.32	0.31	-0.34	0.50	0.57	0.56	0.17
Homicides	-0.31	0.09	-0.04	0.11	-0.07	-0.24	-0.26	-0.21

Table 9: Correlation Matrix: Columns 25 to 28

	FoodInsecurityPct	RuralPopulGrowth	VoiceAccountability	Homicides
HDI_Index	-0.83	-0.24	0.59	-0.24
IHDI_Index	-0.85	-0.52	0.65	-0.30
GiniCoeff	0.84	0.50	-0.58	0.26
FoodIndex	0.14	-0.02	0.10	-0.01
CorruptionScore	-0.62	-0.30	0.77	-0.24
ElectricAccess	-0.78	-0.23	0.40	-0.06
PopDensity	-0.13	-0.03	0.08	-0.09
PopInSlums	0.77	0.48	-0.55	0.17
WaterStress	-0.08	-0.13	-0.14	-0.09
BankingAccess	-0.69	-0.44	0.65	-0.28
RnDInvest	-0.49	-0.05	0.54	-0.31
GovtEffectiveness	-0.74	-0.31	0.75	-0.24
ShippingIndex	-0.47	-0.24	0.13	-0.23
FixedIntSubs	-0.72	-0.12	0.54	-0.22
InternetUsersPct	-0.77	-0.16	0.44	-0.20
PoliticalStability	-0.51	-0.31	0.68	-0.18
RuleOfLaw	-0.68	-0.31	0.82	-0.31
RqdEduYears	-0.30	-0.12	0.32	0.09
GovtEduSpendPctGDP	-0.22	-0.08	0.31	-0.04
MortalityFromDirtyness	0.70	0.49	-0.34	0.11
UHCServiceCoverage	-0.79	-0.46	0.50	-0.07
HealthSpendPerCapita	-0.52	-0.10	0.57	-0.24
GovtHealthSpendPerCapita	-0.53	-0.11	0.56	-0.26
MigrantsPct	-0.39	-0.05	0.17	-0.21
FoodInsecurityPct	1.00	0.43	-0.54	0.39
RuralPopulGrowth	0.43	1.00	-0.31	0.01
VoiceAccountability	-0.54	-0.31	1.00	-0.04
Homicides	0.39	0.01	-0.04	1.00

Figure 1

COMMENTS:

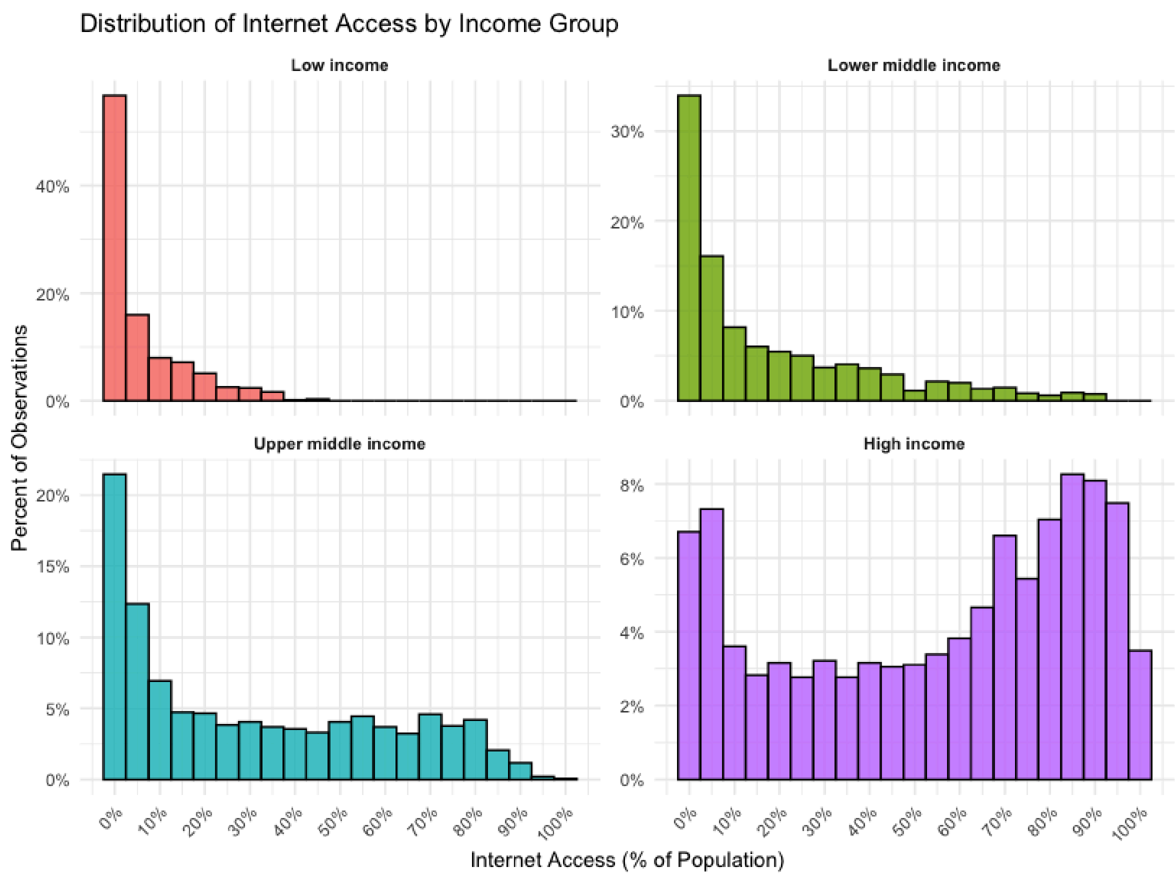
- This scatterplot shows digital inclusion versus our wellbeing measure, with a trendline and R^2
- The positively sloped trendline shows that our basic hypothesis has promise as it does show correlation
- However, the low R^2 supports our prediction that predictability of wellbeing based on internet access alone is unlikely due to other factors

Figure 2

COMMENTS:

- This scatterplot shows the same relationship, but separated into income group cohorts
- We can observe the following:
  - The lower the income of the country cohort, the stronger the correlation between internet access and wellbeing (steeper trend line)
  - However, predictability is tied to income - the R^2 for lower income groups is much lower, suggesting (logically) that lower income countries have far more variables impacting wellbeing than higher income countries
- This plot supports the idea of clustering and / or controlling by income group - we'll see more on this later
- This plot and the previous both seem to show that much of the data (likely in earlier years when the internet was new) is clustered at around 0% internet access. We'll explore this further.

Figure 3



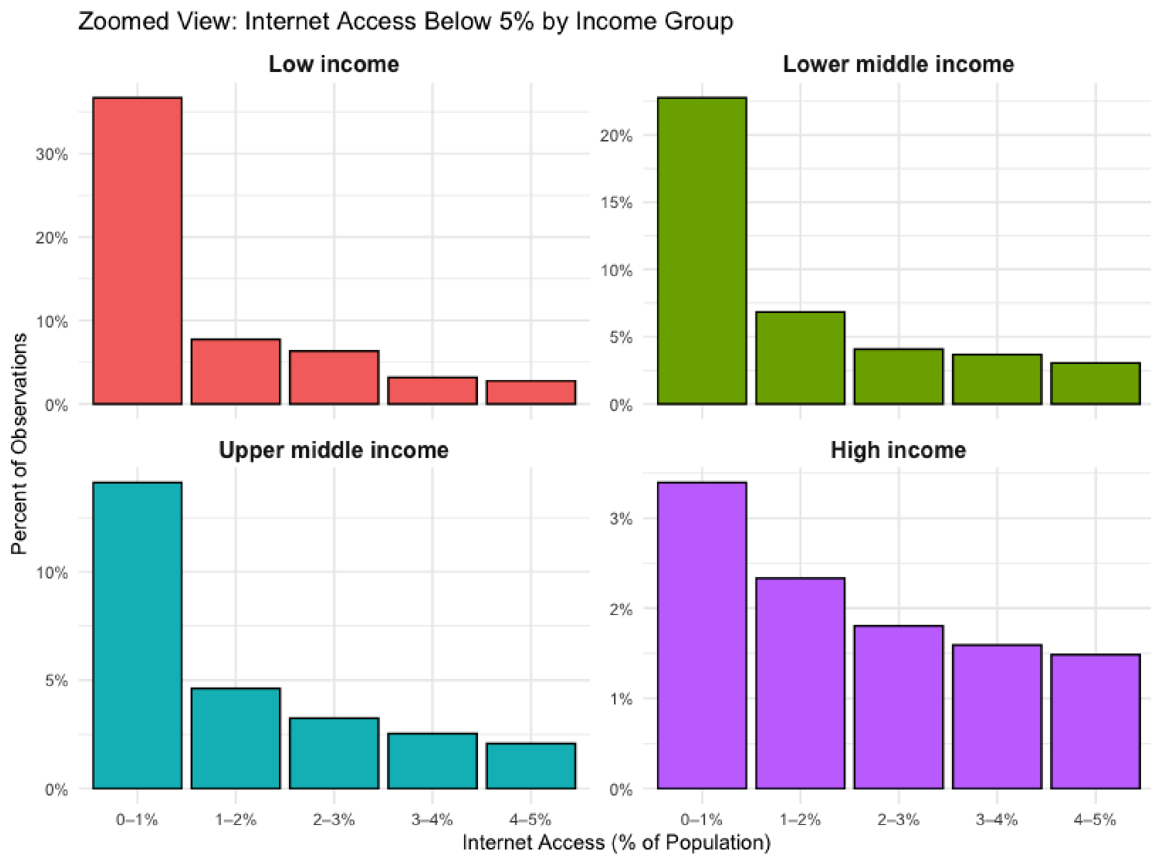
COMMENTS:

- These histograms show that we do indeed have a left skew issue, with data clustered near the 0% internet access side of the X axis

- However, what seems quite evident is that this problem is far more concentrated in low and middle income countries, which presumably were much later to the "internet party"

- From a modeling perspective, it's useful to see how many of these data points are literally 0% since this left skew at 0% internet access may introduce considerable variance into our predictive models (which we see on the trendlines in the previous plots)

Figure 4



COMMENTS:

- Indeed, if we "explode" just the 0%-5% part of the plot, we do in fact observe that for all but the High Income countries, internet access of less than 1% (effectively 0) represents a very significant part of the data set

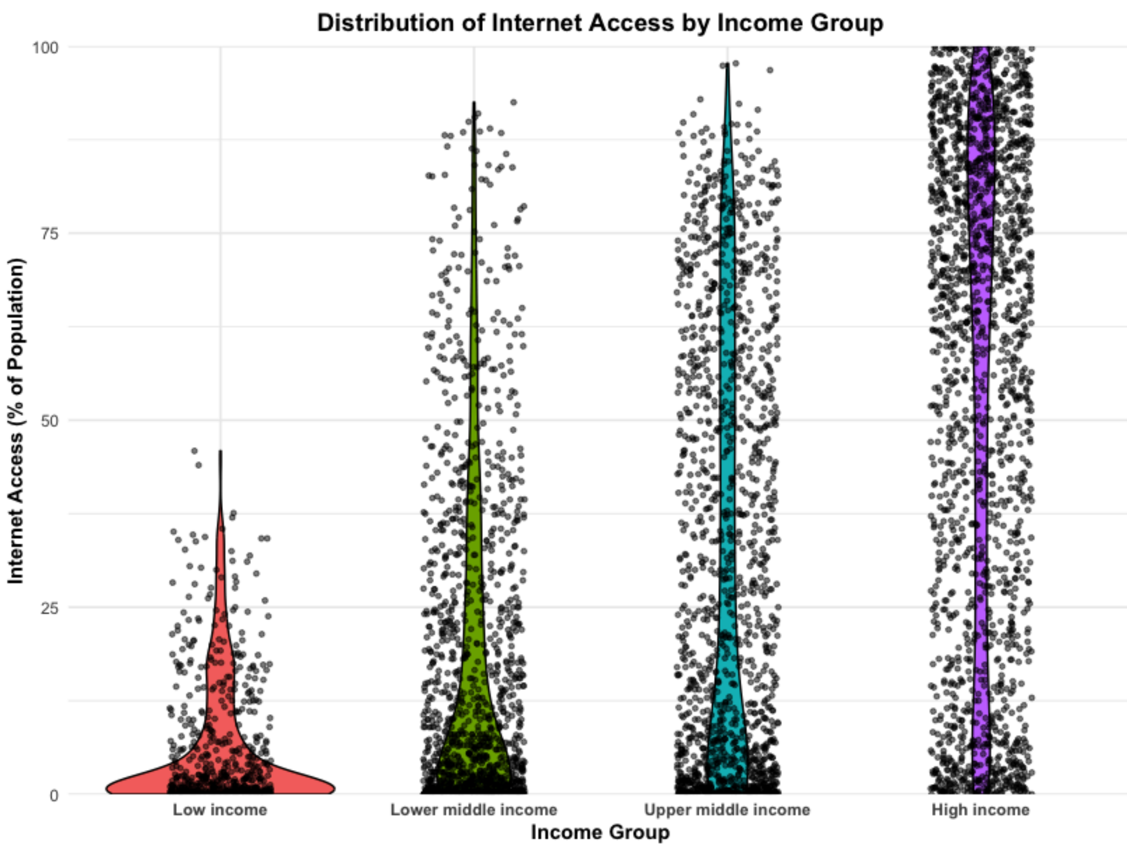
- This will need to be dealt with in our modeling approach to avoid massive variance caused by this data skew

Figure 5

COMMENTS:

- This violin plot shows the extent to which wellbeing differs based on income (which is one of the three components of the HDI index we use to measure wellbeing
- Not surprisingly, higher income groups show a higher overall level of wellbeing
- What we also see is that the distribution of HDI within each income group differs:
    - For the two lowest income categories, HDI can vary widely across its band relative to its median
    - The upper middle income group sees the variability in HDI decline, and still equally distributed around its median income
    - The High Income group's HDI band shows numerous outliers, skewed NEGATIVELY towards lower HDI which suggests that at higher overall national income levels, the impact of other factors like education and health can only worsen wellbeing, not improve it

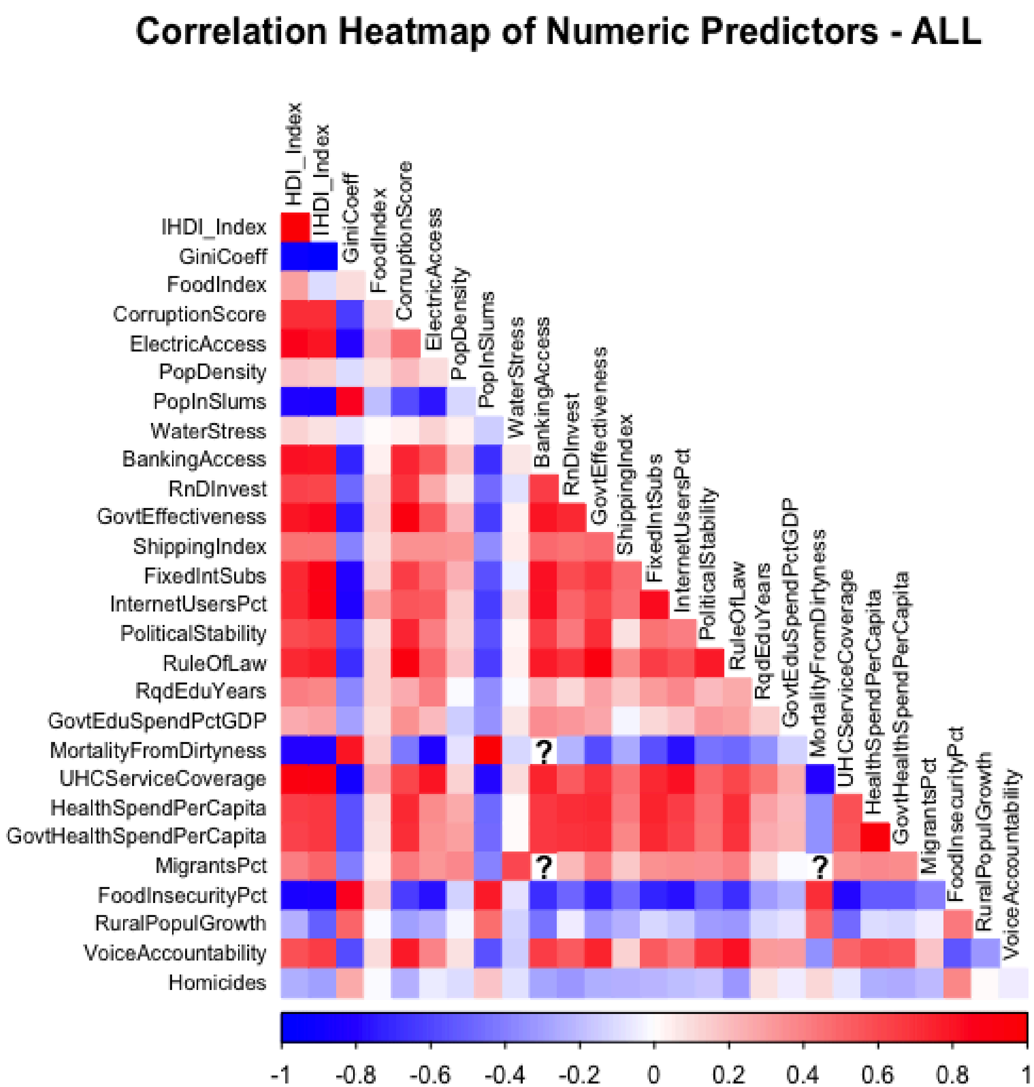
Figure 6



COMMENTS:

- This violin plot shows just how different internet access is based on income grouping
- Bearing in mind that the data analyzed contains time series, we can conclude:
  - Low income countries generally have shown slow if any progress at expanding access to the internet
  - Among wealthier nations, the violin plots get taller, thinner and more dispersed, representing the positive relationship between a country's income (and therefore its wellbeing) and its digital inclusion
  - HOWEVER- whether correlation = causation remains to be seen - could it simply be that citizens of wealthier countries simply are more likely to be able to afford internet access, and we have it backwards?

Figure 7



COMMENTS:

- This heatmap displays a comprehensive overview of the correlation coefficients for all numeric predictor variables in the dataset.
- A color gradient is used to represent the strength and direction of correlations. Dark blue indicates strong negative correlation, white indicates no correlation, and dark red indicates strong positive correlation.
- Several instances of missing correlation values are noted by question marks that suggest incomplete data for those specific variable pairs.
- This type of heatmap is valuable for understanding the interdependencies within a dataset and for identifying potential multicollinearity as well as features that might be good candidates for feature engineering.

Figure 8

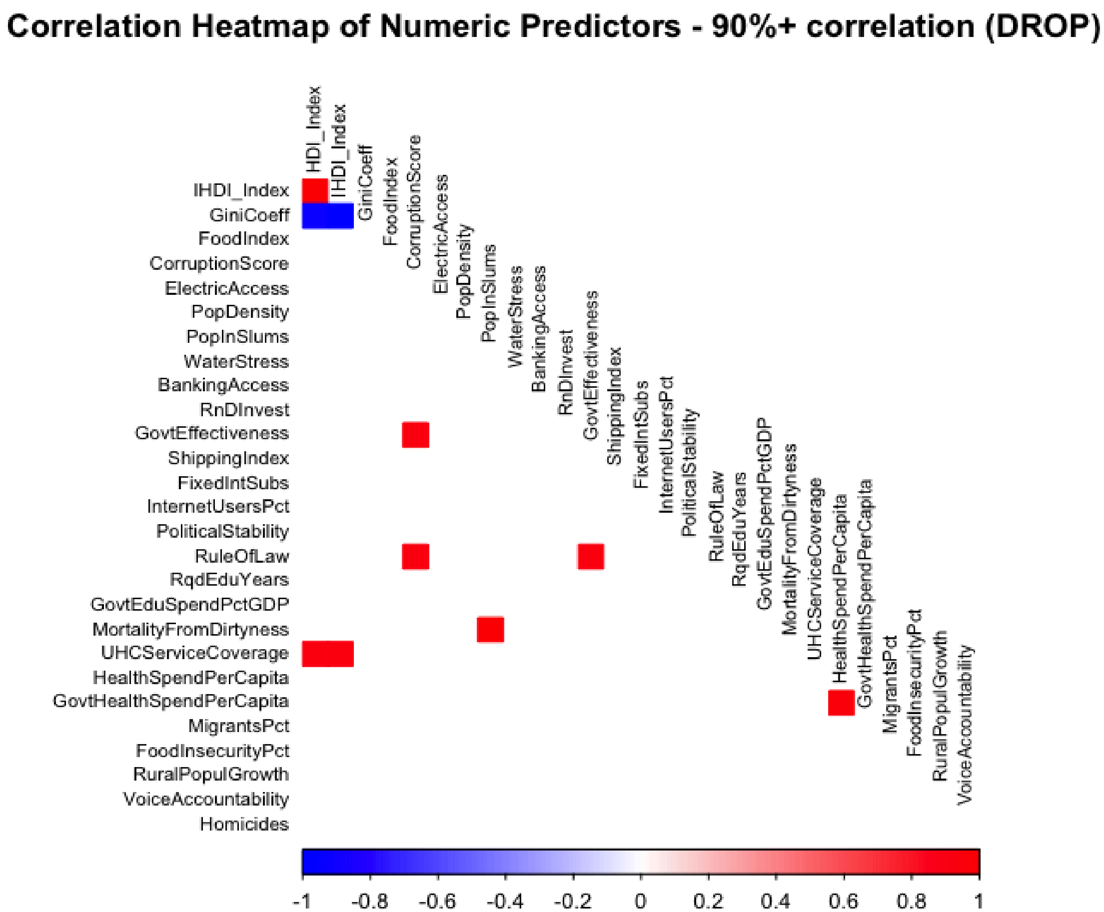
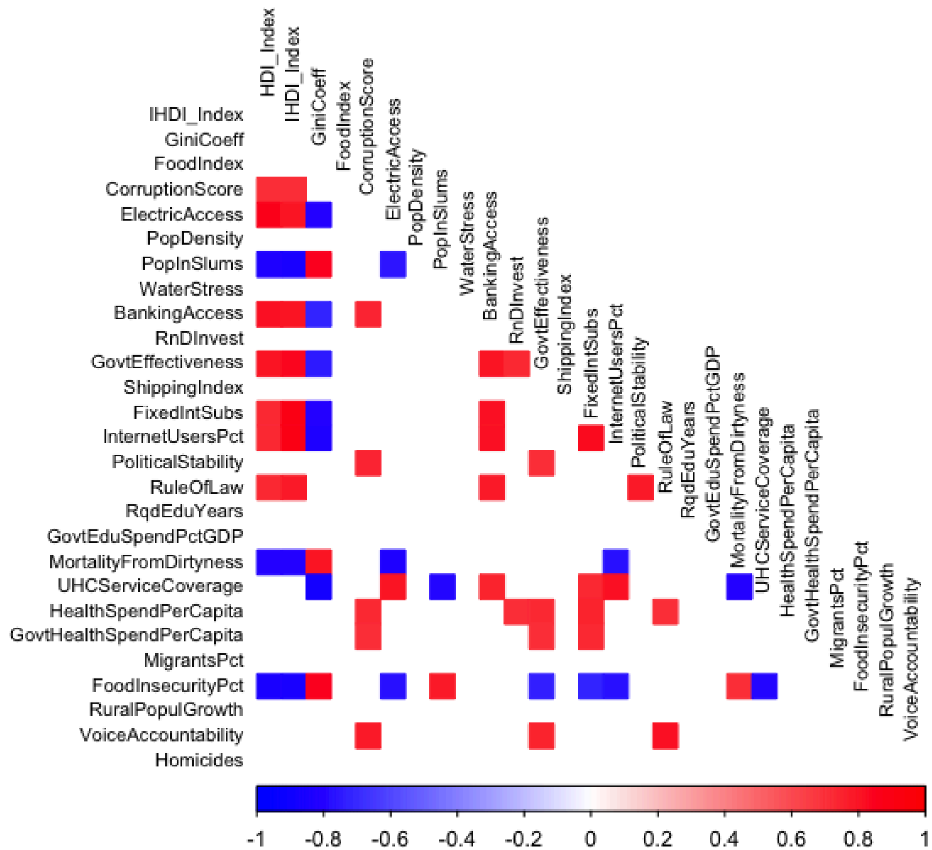


Figure 9

COMMENTS:

- This chart is the same correlation heatmap filtered to highlight numeric predictor variables with a correlation of 90% or higher.
- Correlations between features (NOT responses) of 90% or greater should lead to dropping one of the features in statistical models due to collinearity concerns
- Strong positive correlations (dark red) are evident, for example, between "IHDI_Index" and "HDI Index" as well as between both of these indexes and "UHCServiceCoverage"

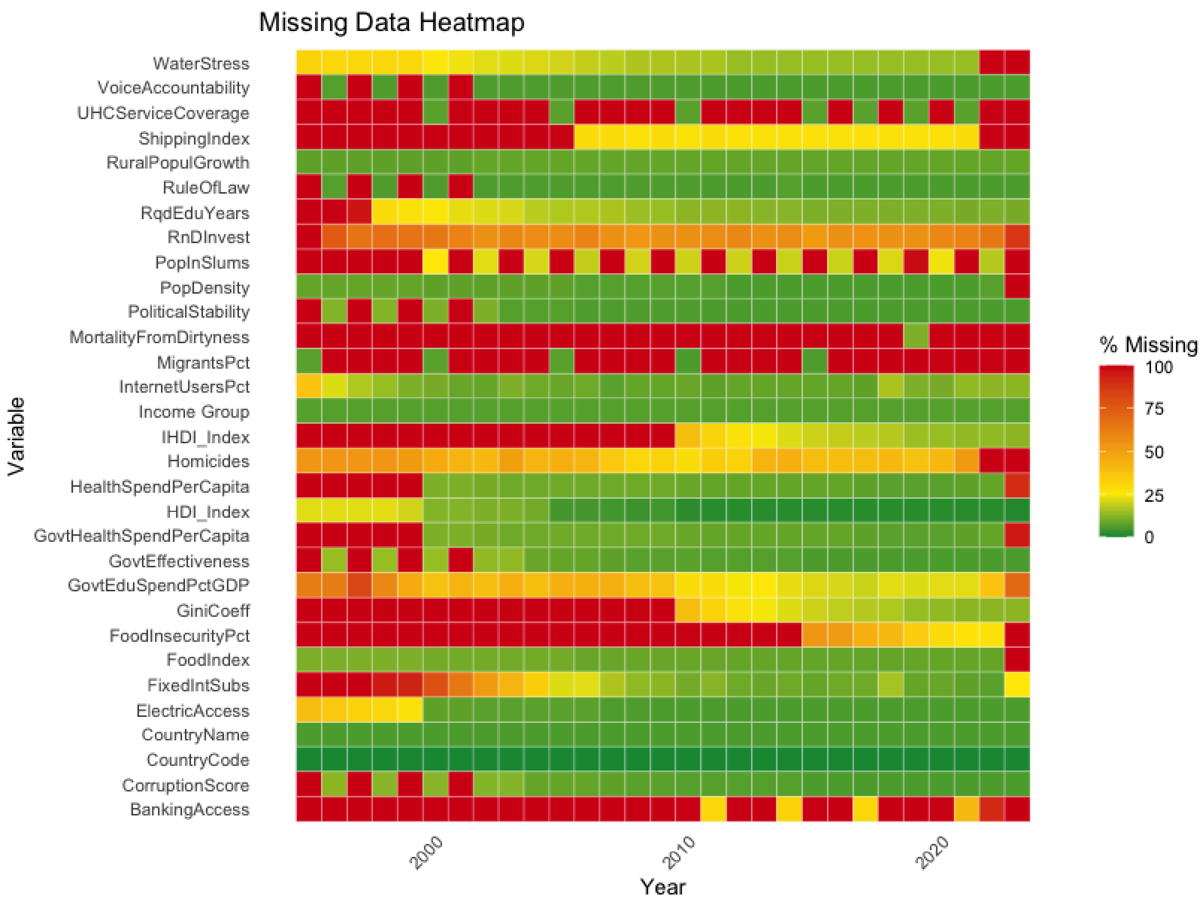
Correlation Heatmap of Numeric Predictors - 70%-90%+ correlation (VIF)



COMMENTS:

- This version filters variable pairs with correlations between 70% and 90%
- If used in statistical models, these features should be examined using VIF analysis to determine how much their collinearity may impact variance, and which (if any) should be dropped

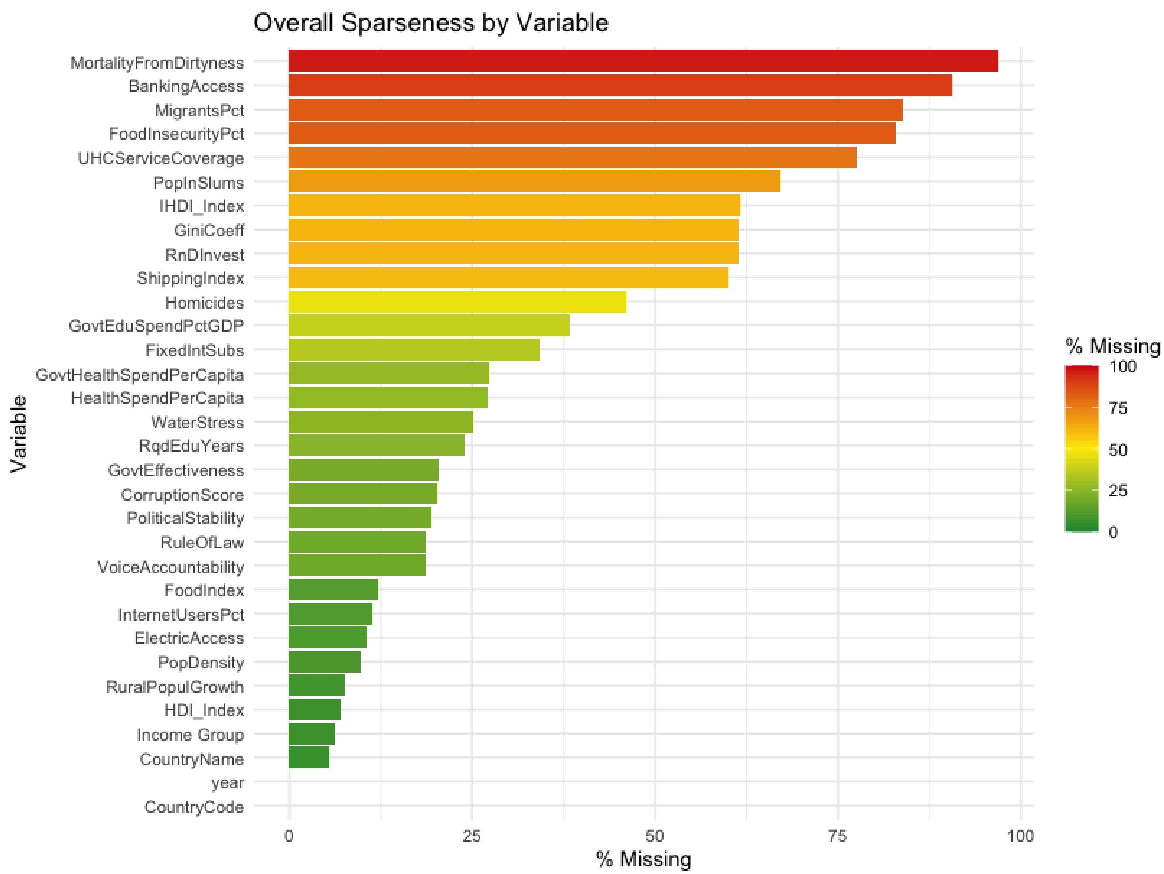
Figure 10



COMMENTS:

- This heatmap chart exhibits the amount of missing data per feature (vertically) and over each year covered by our study (horizontally).
- There are several features that have a large amount of data missing over the complete range of years covered by this study such as "PopInSlums" and "MortalityFromDirtyness"
- Generally the amount of data becomes more complete as the data collection period reaches more current dates.

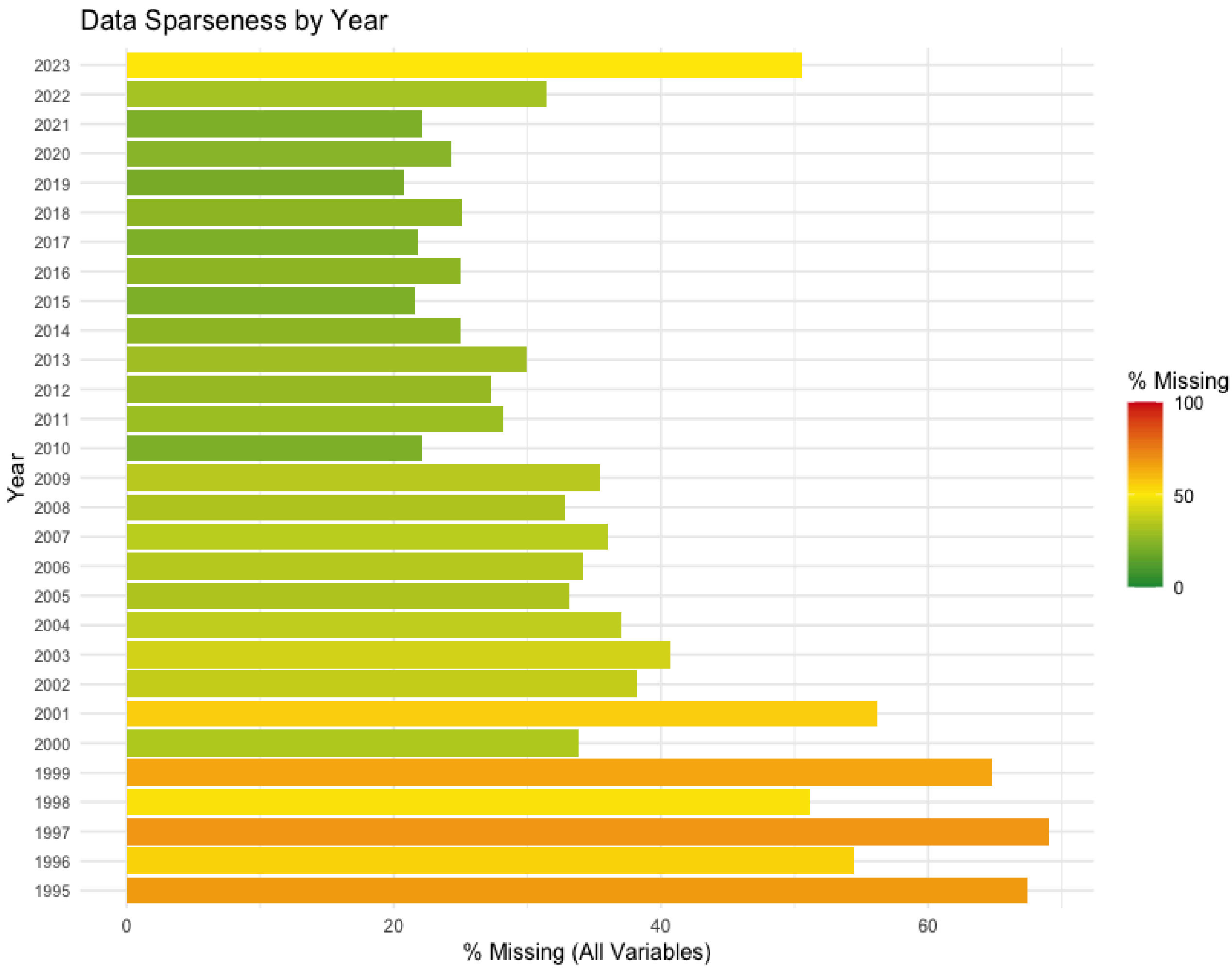
Figure 11



COMMENTS:

- This exhibit visualizes the sparseness of each variable as an indicator of which variables may warrant exclusion from the upcoming steps and which ones will need less imputation.
- This chart serves as an excellent overview of the nature of completeness of each variable, but it is important to take into consideration factors like the GiniCoefficient which is not collected each year that the data range accounts for.

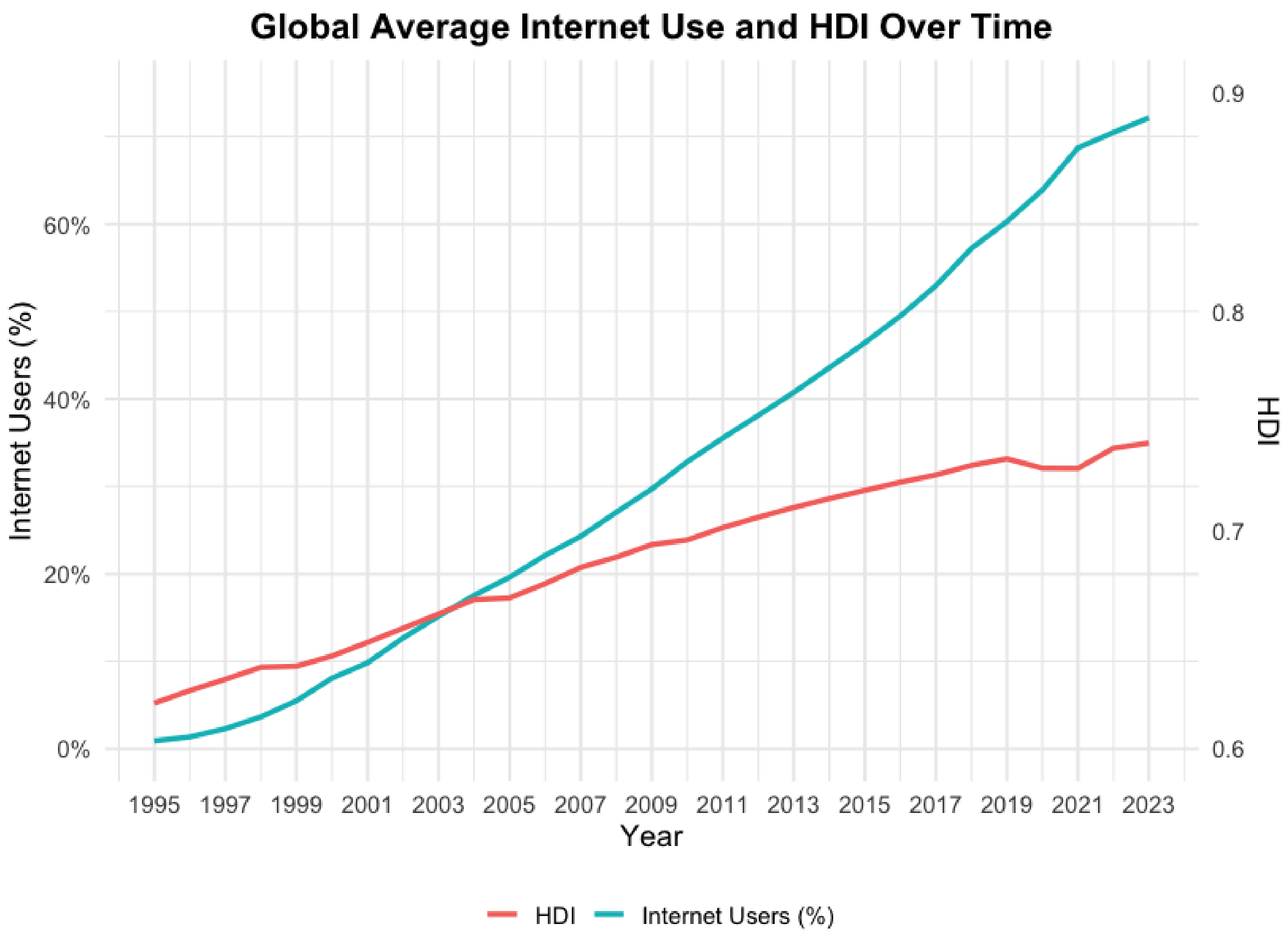
Figure 12



Comments:

- This chart visualizes the percentage of missing data across all variables for each year from 1995 to 2023.
- Earlier years exhibit the highest data sparseness, along with 2023 (the most recent year), with a significant portion of variables missing which is indicated by yellow-orange bars exceeding 50% missingness.
- Data completeness generally improves over time from 2004 onwards and shows considerably less missing data, due to broader collection of data globally by organizations.

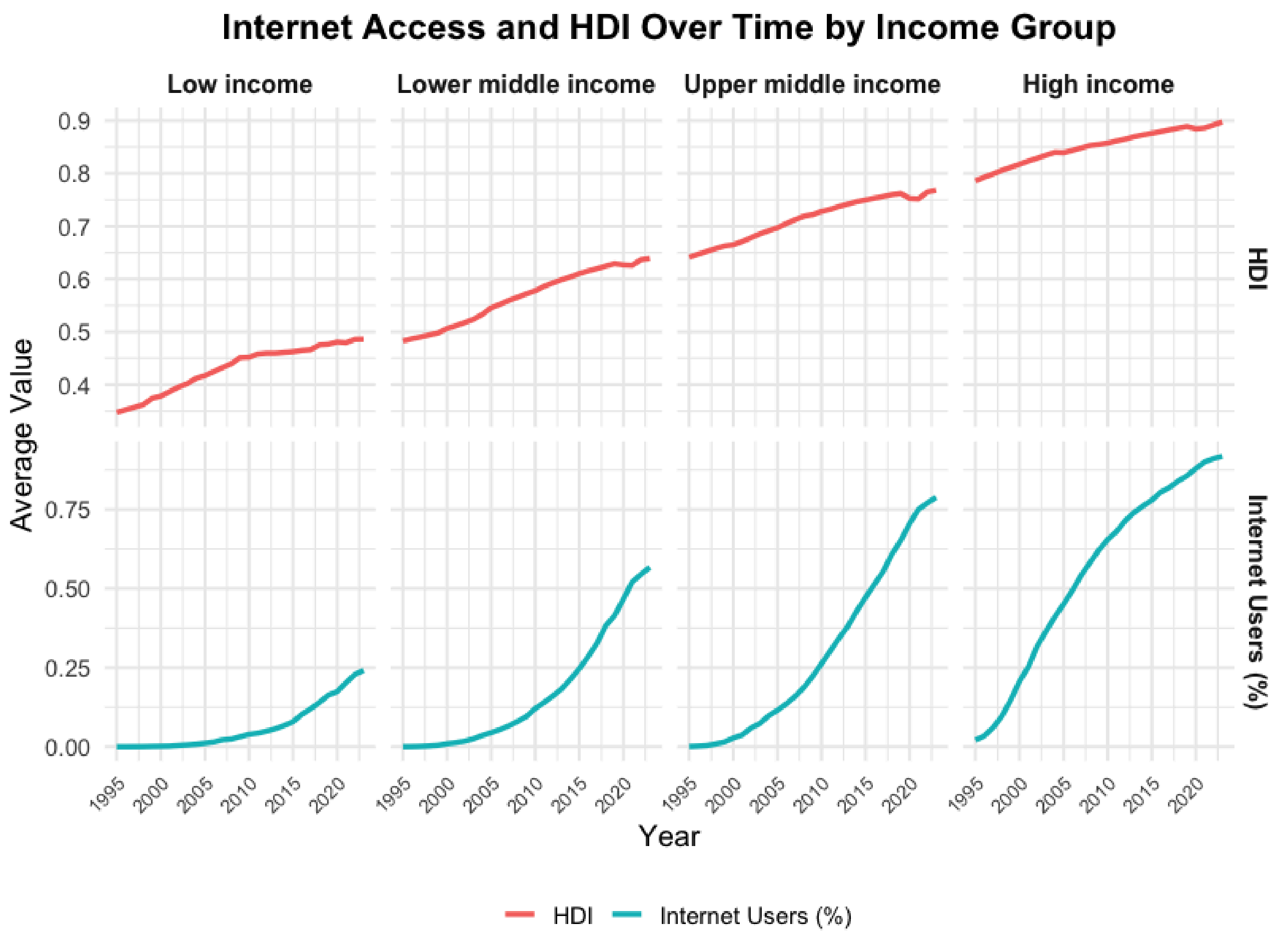
Figure 13



Comments:

- This chart visualizes the growth in average GLOBAL internet usage and HDI over time
- The relatively linear growth trajectories of both bode well for a predictive model correlating the two

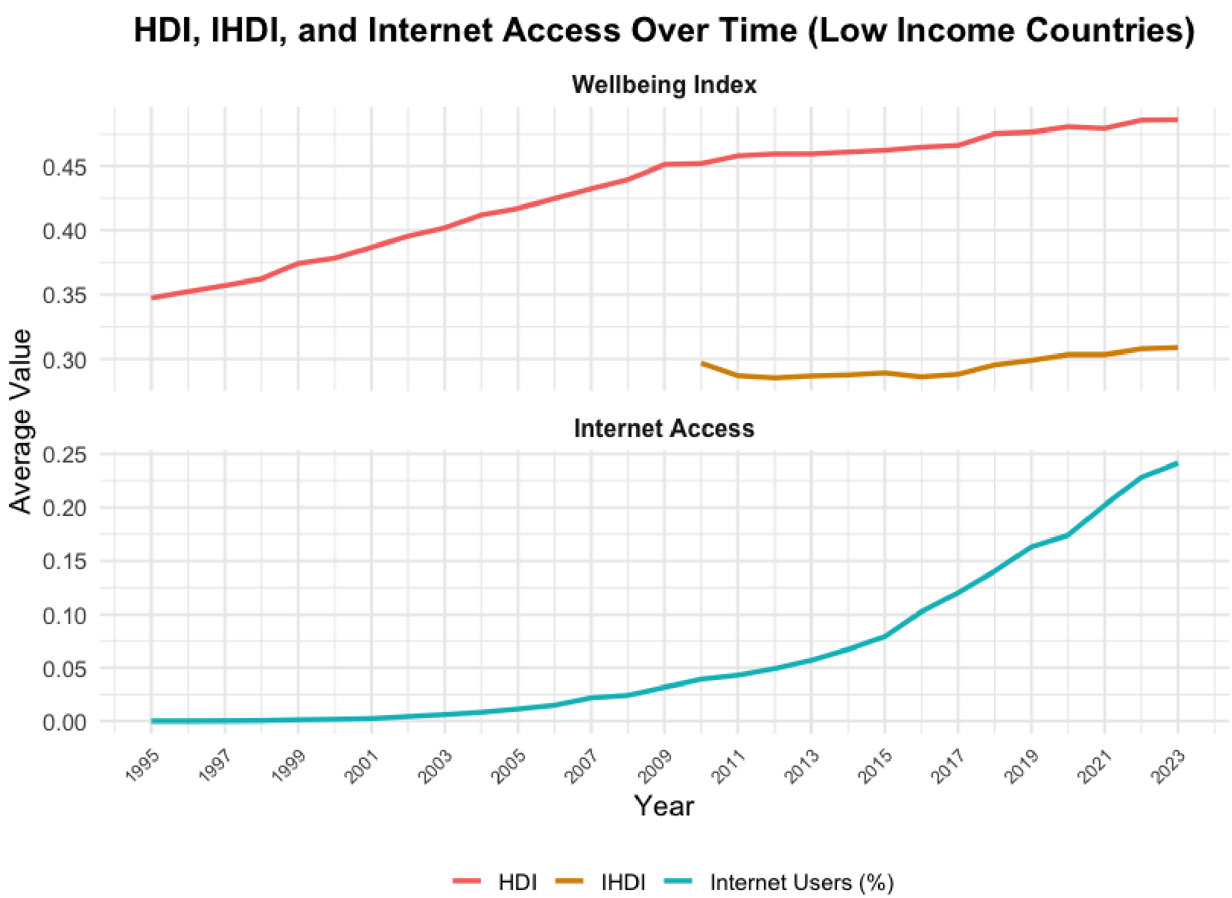
Figure 14



Comments:

- This chart breaks this data out by Income Group Cohort
- It's clear how much HDI is related to income given the differences in HDI across the income groups
- The higher the income cohort, the sooner internet access growth began and the steeper the adoption curve (good for our thesis)
- HOWEVER - there seems to be a flattening of the HDI growth trajectory for Low Income countries during the same period that internet usage growth finally starts to occur!

Figure 15



Comments:

- Indeed - that flattening is noticeable
- Could this actually REFUTE our thesis, suggesting that internet access for Low Income countries is actually a DRAG on wellbeing?
- Or could it be that either the global recession (which disproportionately impacted these countries) offset the impact of internet access….
- …or that the growing impact of income inequality shown on this chart means the positive impact of internet access did not impact everyone equally?

Appendix C

XGBoost Model Predicting the HDI Index

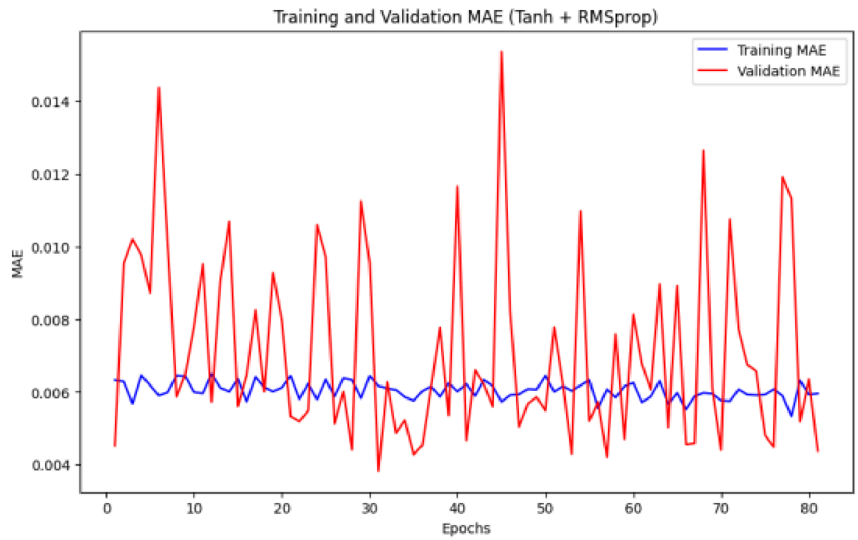
Iteration and Cross Validation Results

eta	max_depth	roundsrun	subsample	rmse	mae	best_iteration	run_id
0.01	6	682	0.6	0.0421	0.0320	672	1
0.01	6	687	0.8	0.0423	0.0323	677	10
0.01	18	949	0.6	0.0426	0.0324	939	7
0.01	12	866	0.6	0.0426	0.0324	856	4
0.10	6	81	0.6	0.0427	0.0326	71	2
0.01	12	978	0.8	0.0428	0.0327	968	13
0.01	18	1114	0.8	0.0428	0.0327	1104	16
0.01	6	601	1.0	0.0429	0.0326	591	19
0.01	18	1178	1.0	0.0431	0.0326	1168	25
0.01	12	1181	1.0	0.0431	0.0327	1171	22
0.10	6	100	0.8	0.0431	0.0327	90	11
0.10	18	124	0.6	0.0433	0.0330	114	8
0.10	12	118	0.6	0.0435	0.0332	108	5
0.10	12	108	0.8	0.0439	0.0337	98	14
0.10	6	93	1.0	0.0439	0.0334	83	20
0.10	18	163	0.8	0.0440	0.0337	153	17
0.10	18	145	1.0	0.0442	0.0337	135	26
0.10	12	145	1.0	0.0442	0.0338	135	23
0.30	6	31	1.0	0.0445	0.0339	21	21
0.30	6	30	0.6	0.0450	0.0346	20	3
0.30	12	64	1.0	0.0466	0.0356	54	24
0.30	18	49	0.6	0.0467	0.0357	39	9
0.30	18	64	1.0	0.0468	0.0355	54	27
0.30	12	56	0.6	0.0469	0.0356	46	6
0.30	6	29	0.8	0.0470	0.0358	19	12
0.30	12	35	0.8	0.0478	0.0362	25	15
0.30	18	44	0.8	0.0479	0.0365	34	18

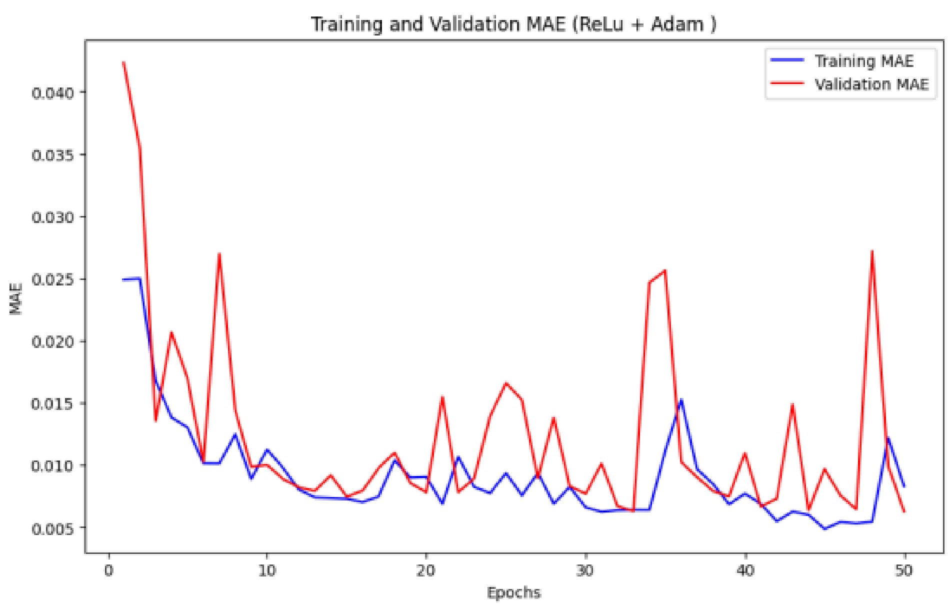
Appendix D

Feed-Forward Neural Network Predicting HDI Index

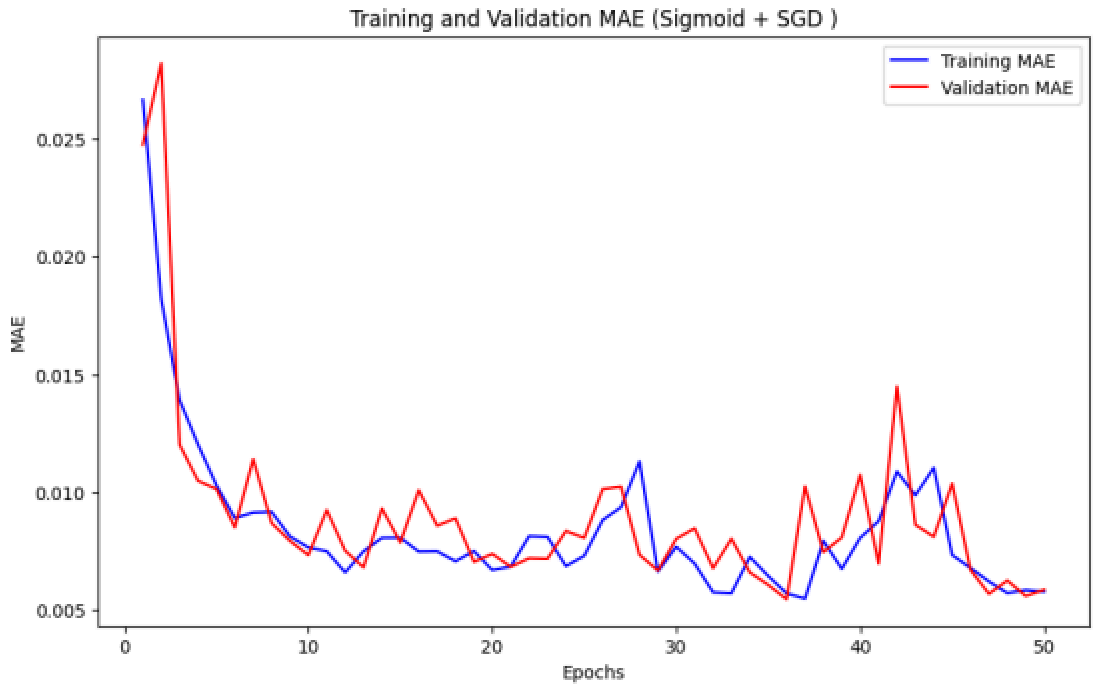
Training and Validation Errors Across Iterations



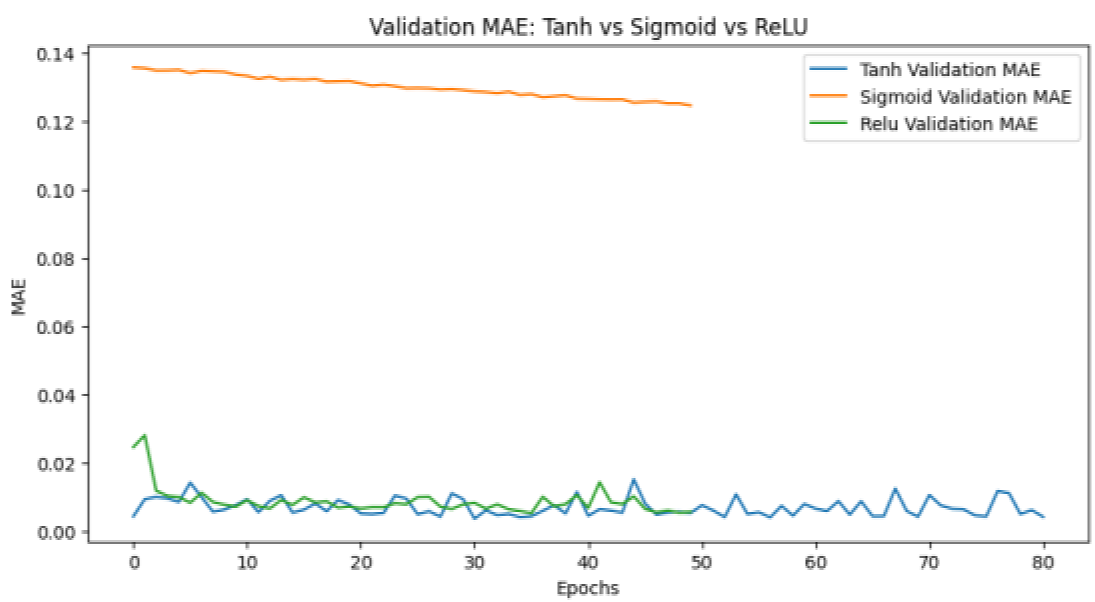
Training and Validation MAE performance over 80 epochs: FFNN - Tanh/RMSProp



Training and Validation MAE performance over 50 epochs: FFNN - ReLu/Adam



Training and Validation MAE performance over 50 epochs: FFNN - Sigmoid/SGD



Comparison of MAE Validation Performance across all three models

Appendix E

XGBoost Model Predicting the HDI Category

Iteration and Cross Validation Result

eta	max_depth	roundsrun	subsample	loglossmean	accuracy	best_iteration	run_id
0.10	6	39	0.6	0.6156	0.7744	29	2
0.30	6	23	0.8	0.6205	0.7654	13	12
0.10	18	47	0.8	0.6322	0.7582	37	17
0.30	6	23	1.0	0.6375	0.7498	13	21
0.10	6	31	0.8	0.6585	0.7680	21	11
0.30	6	35	0.6	0.6786	0.7578	25	3
0.30	18	38	0.8	0.6876	0.7604	28	18
0.30	12	36	0.6	0.7028	0.7500	26	6
0.10	12	28	0.6	0.7039	0.7524	18	5
0.30	18	42	0.6	0.7070	0.7491	32	9
0.30	12	46	0.8	0.7198	0.7514	36	15
0.30	12	35	1.0	0.7298	0.7484	25	24
0.30	18	39	1.0	0.7337	0.7420	29	27
0.10	18	23	1.0	0.7904	0.7526	13	26
0.10	12	22	1.0	0.8015	0.7506	12	23
0.10	18	21	0.6	0.8122	0.7702	11	8
0.10	12	19	0.8	0.8778	0.7498	9	14
0.10	6	16	1.0	0.9633	0.7573	6	20
0.01	18	50	0.8	1.0762	0.7584	40	16
0.01	18	42	0.6	1.1216	0.7646	32	7
0.01	6	24	0.6	1.2551	0.7586	14	1
0.01	18	23	1.0	1.2620	0.7415	13	25
0.01	6	23	1.0	1.2623	0.7545	13	19
0.01	6	22	0.8	1.2694	0.7622	12	10
0.01	12	19	0.6	1.2976	0.7563	9	4
0.01	12	18	0.8	1.3070	0.7475	8	13
0.01	12	14	1.0	1.3445	0.7469	4	22